Chapter 7

Instruments and Equipment for Tooth Preparation

"A man who works with his hands is a … **Labourer**
A man who works with hands and his brain is a … **Craftsman**
A man who works with his hands and his brain and his heart is an … **Artist***".*

—Louis Nizer

Removal and shaping of tooth structures are essential aspects of restorative dentistry. Initially, this was a difficult process accomplished entirely by the use of hand instruments. The introduction of rotary-powered cutting equipment was a major advancement in dentistry. From the time of the first hand-powered dental drill to the present-day air-driven and electric handpiece, tremendous strides have been made in the mechanical alteration of tooth structure and in the ease with which teeth may be restored. Modern high-speed equipment has eliminated the need for many hand instruments for tooth preparation. Nevertheless, hand instruments remain an essential part of the armamentarium for restorative dentistry.

Hand Instruments for Cutting

The early hand-operated instruments—with their large, heavy handles (Fig. 7.1) and inferior (by present standards) metal alloys in the blades—were cumbersome, awkward to use and ineffective in many situations. As the commercial manufacture of hand instruments increased and dentists began to express ideas about tooth preparation, it became apparent that some scheme for identifying these instruments was necessary. Among his many contributions to modern dentistry, G.V. Black is credited with the first acceptable nomenclature for and classification of hand instruments.[1] His classification system enabled dentists and manufacturers to communicate more clearly and effectively about instrument design and function.

Modern hand instruments, when properly used, produce beneficial results for the operator and the patient. Some results may only be satisfactorily achieved with hand instruments (i.e. not with rotary instruments). Preparation form dictates some circumstances in which hand instruments are to be used, whereas accessibility dictates others.

Terminology and Classification

Categories

The hand instruments used in the dental operatory may be categorized as shown in Box 7.1.[1]

Design

Most hand instruments, regardless of their use, are composed of three parts—blade, shank and handle (Fig. 7.2).

Blade

The blade is the working end of the instrument and is connected to the handle by the shank.

For many non-cutting instruments, the part corresponding to the blade is termed as *nib*.

The end of the nib, or working surface, is known as *face*.

Some instruments have a blade on both ends of the handle and are known as double-ended instruments. The blades are of many designs and sizes, depending on their functions.

Shank

Shanks, which serve to connect the handles to the working ends of the instruments, are normally smooth, round and tapered. They often have one or more bends to overcome the tendency of the instrument to twist while in use when force is applied.

The functional orientation and length of the blade determine the number of angles in the shank. Black classified instruments on the basis of the number of shank angles as:



Figure 7.1 Designs of some early hand instruments. These instruments were individually handmade, variable in design and cumbersome to use. Because of the nature of the handles, effective sterilization was a problem.

- Mon-angle (one)
- Bin-angle (two)
- Triple-angle (three)[2]

Instruments with small, short blades may be easily designed in mon-angle form while confining the cutting edge within the required limit. Instruments with longer blades or more complex orientations may require two or three angles in the shank to bring the cutting edge close to the long axis of the handle. Such shanks are termed *contra-angled*.

Box 7.1 Classification of hand instruments

Cutting hand instruments:
- Excavators
- Ordinary hatchets
- Hoes
- Angle formers
- Spoons
- Chisels
- Straight chisels
- Curved chisels
- Bin-angle chisels
- Enamel hatchets
- Gingival margin trimmers

Other cutting instruments:
- Knives
- Files
- Scalers
- Carvers

Non-cutting hand instruments:
- Amalgam condensers
- Mirrors
- Explorers
- Probes

Handle

I. Handles are available in various sizes and shapes. Most instrument handles are small in diameter (5.5 mm) and light.
II. They are commonly eight-sided and knurled to facilitate control.

Balance

Definition. Balance is an attribute that allows for the concentration of force onto the blade without causing rotation of the instrument in the operator's grasp.

Enamel and dentin are difficult substances to cut and require the generation of substantial forces at the tip of the instrument. Therefore, cutting hand instruments must be balanced and sharp.

Clinical Notes

- Balance is accomplished by designing the angles of the shank so that the cutting edge of the blade lies within the projected diameter of the handle and nearly coincides with the projected axis of the handle (Fig. 7.3; Fig. 7.2).
- For optimal anti-rotational design, the working end of the instrument is brought within 2 mm to the long axis of the handle.
- All dental instruments and equipment need to satisfy this principle of balance.

Names

Black classified all of the instruments by name. In addition, for hand-cutting instruments, he developed a numeric formula to characterize the dimensions and angles of the working end (see the next section for details of the formula). Black's classification system by instrument name categorized instruments by:

I. function (e.g. scaler and excavator),
II. manner of use (e.g. hand condenser),

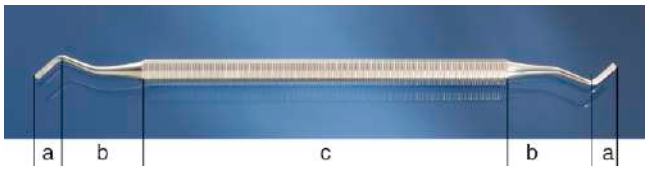


Figure 7.2 Double-ended instrument illustrating three component parts of hand instruments: blade (*a*), shank (*b*) and handle (*c*). (Modified from Boyd LRB: *Dental instruments: a pocket guide*, ed 4, St. Louis, MO, 2012, Saunders.)

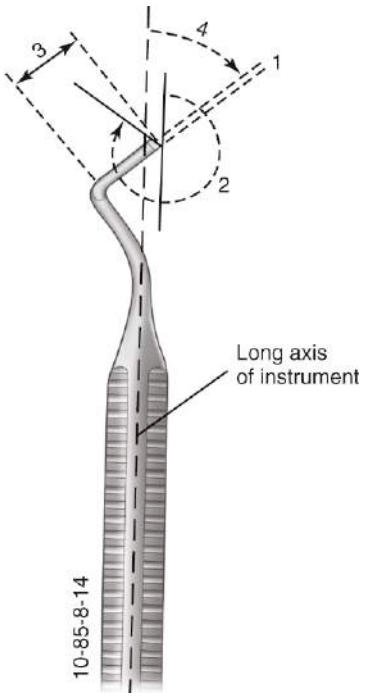


Figure 7.3 Instrument shank and blade design (with primary cutting edge positioned close to handle axis to aid in limiting unwanted instrument rotation). The complete instrument formula (four numbers) is expressed as the blade width (1) in 0.1-mm increments, cutting edge angle (2) in centigrades, blade length (3) in millimetres and blade angle (4) in centigrades.

III. design of the working end (e.g. spoon excavator and sickle scaler) or
IV. shape of the shank (e.g. mon-angle, bin-angle, contra-angle).[2]

These names were combined to form the complete description of the instrument (e.g. bin-angle spoon excavator).

Formulas

Cutting instruments have formulas describing the dimensions and angles of the working end. These are placed on the handle using a code of three or four numbers separated by dashes or spaces (e.g. 10–85–8–14) (Fig. 7.4; Fig. 7.3).

First number: The first number indicates the width of the blade or primary cutting edge in tenths of a millimetre (0.1 mm) (e.g. 10 = 1 mm).

Second number: The second number of a four-number code indicates the primary cutting edge angle, measured from a line parallel to the long axis of the instrument handle in clockwise centigrade. The centigrade angle is expressed as a percent of 360 degrees (e.g. 85 = 85% × 360 degrees = 306 degrees). The instrument is positioned so that this number always exceeds 50. If the edge is locally perpendicular to the blade, this number is normally omitted, resulting in a three-number code (Fig. 7.4).



Figure 7.4 Examples of instruments with three-number and four-number instrument formulas.

Third number: The third number (second number of a three-number code) indicates the blade length in millimetres (e.g. 8 = 8 mm).

Fourth number: The fourth number (third number of a three-number code) indicates the blade angle, relative to the long axis of the handle in clockwise centigrade (e.g. 14 = 50.4 degrees). The instrument is positioned such that this number is always 50 or less. The most commonly used hand instruments, including those specified in this text, are shown in Figs. 7.5–7.9.

In some instances, an additional number on the handle is the manufacturer's identification number (Fig. 7.4). Identification numbers are included to assist specific manufacturers in cataloguing and should not be confused with formula numbers.

Bevels

Most hand-cutting instruments have a single bevel on the end of the blade that forms the *primary cutting edge*. Two additional bevels form *secondary cutting edges* and extend from the primary edge for the length of the blade

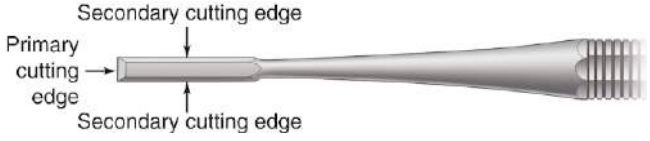


Figure 7.5 Chisel blade design showing primary and secondary cutting edges.

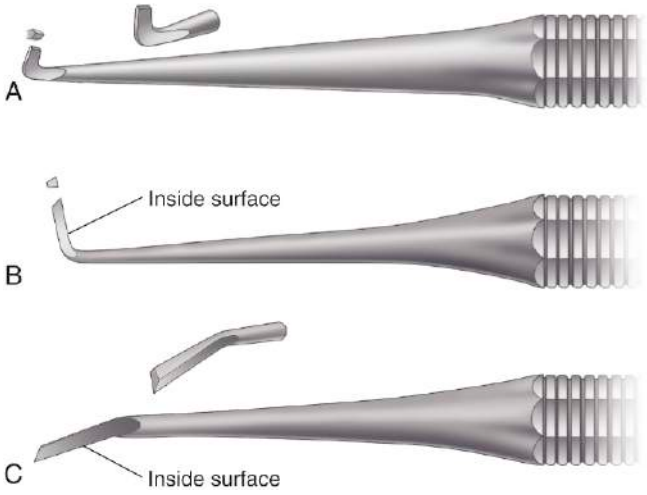


Figure 7.6 Examples of hand instruments called excavators. (A) Bi-bevelled ordinary hatchet. (B) Hoe with distal bevel. (C) Angle former with distal bevel.

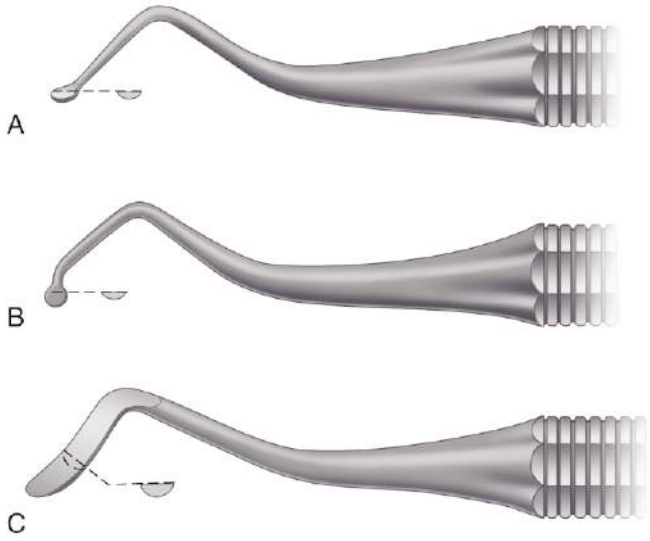


Figure 7.7 Examples of hand instruments called spoon excavators. (A) Bin-angle spoon. (B) Triple-angle spoon. (C) Spoon.

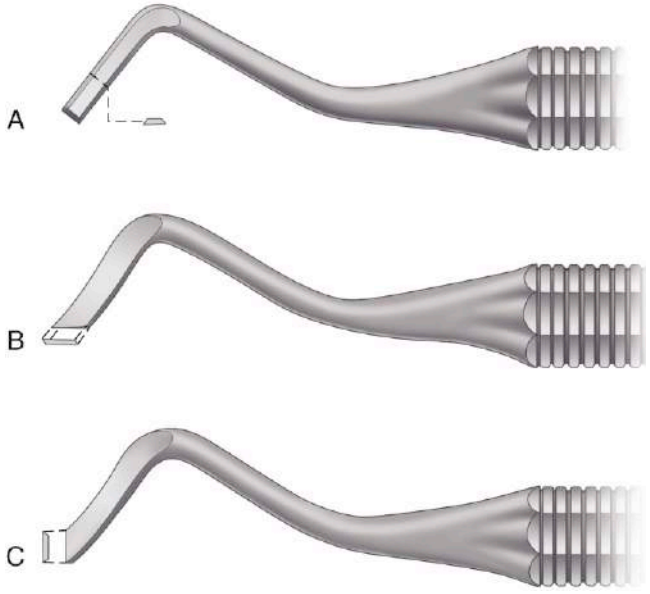


Figure 7.8 Examples of hand instruments called chisels. (A) Enamel hatchet. (B) Gingival margin trimmer. (C) Gingival margin trimmer.

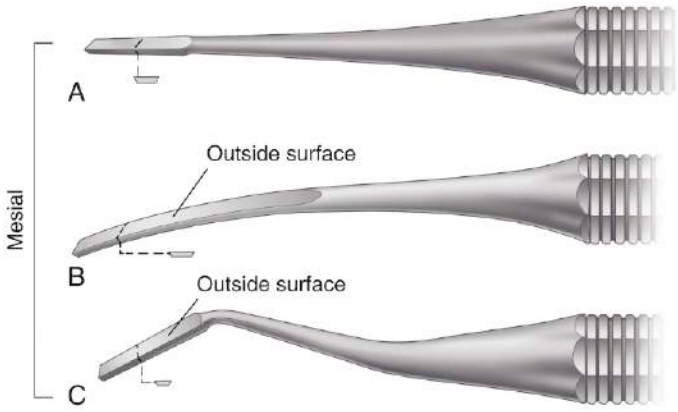


Figure 7.9 Examples of hand instruments called chisels. (A) Straight. (B) Wedelstaedt with mesial bevel. (C) Bin-angle with mesial bevel.

(Fig. 7.5). Bi-bevelled instruments such as ordinary hatchets have two bevels that form the cutting edge (Fig. 7.6A).

Certain single-bevelled instruments, such as *spoon excavators* (Fig. 7.7) and *gingival margin trimmers* (Fig. 7.8B and C), are used with a scraping or lateral cutting motion. *Enamel hatchets* (Fig. 7.8A) may be used with a planning, direct cutting motion and lateral cutting motion. Hatches have the primary cutting edge aligned parallel with the long axis of the handle.

Single-bevelled designs require that the instruments be made in pairs, with the bevels on opposite sides of the blade. These instruments are designated as right bevelled or left bevelled and are indicated by appending the letter R or L to the instrument formula. To determine whether the instrument has a right or left bevel, the primary cutting edge is held down and pointing away and if the bevel appears on the right side of the blade, it is the right instrument of the pair. This instrument, when used in a scraping motion, is moved from right to left. The opposite holds true for the left instrument of the pair. One instrument is suited for work on one side of the preparation and the other is suited for the opposite side of the preparation. The primary cutting edge of the hatchet may be useful for planning the facial and lingual walls of proximal preparations by using a pushing motion and the gingival wall by using a lateral motion. In addition, the secondary cutting edge of the enamel hatchet may be useful for planning the occlusal and gingival walls of posterior Class V preparations.

Most instruments are available with blades and shanks on both ends of the handle. Such instruments are termed *double-ended*. In many cases, the right instrument of the pair is on one end of the handle and the left instrument is on the other end. Occasionally similar blades of different widths are placed on a single double-ended instrument. If one observes the inside of the blade curvature (or the inside of the angle at the junction of the blade and shank) and the primary bevel is not visible, the instrument has a distal bevel (Fig. 7.6). Conversely, if the primary bevel is seen (from the same viewpoint), the instrument has a mesial (or reverse) bevel (Fig. 7.9).

Instruments having the primary cutting edge perpendicular to the axis of the handle are generally termed *chisels* (Fig. 7.9). Examples include bin-angle chisels (Fig. 7.9C), instruments with a slight blade curvature (Wedelstaedt chisels) (Fig. 7.9B) and hoes (Fig. 7.6B), which are single bevelled and not designated as right or left but as having a mesial bevel or a distal bevel. The primary cutting edge of chisels may be useful in planning the occlusal and gingival walls of posterior Class V preparations by using a pushing and/or pulling motion. The primary cutting edge of the hoe is particularly useful for planning the occlusal and gingival walls of Class III preparations. The secondary cutting edge of the hoe is useful for planning the occlusal and gingival walls of Class V preparations as well as walls oriented in an axial direction. Note that hoes may also serve as excavators (see discussion later in text).

Applications

Cutting instruments are used to cut enamel/dentin and restorative materials. Excavators are used for removal of carious tissue, refinement of the internal aspects of the preparation and establishment of correct anatomical restoration form. Chisels are used primarily for cutting enamel.

Excavators

The four subdivisions of excavators are:

I. **Ordinary hatchets:** An ordinary hatchet excavator has the cutting edge of the blade directed in the same plane as that of the long axis of the handle and is bi-bevelled (Fig. 7.6A). These instruments are used primarily on anterior teeth for preparing retentive areas and sharpening internal line angles, particularly in preparation for direct gold restorations.

II. **Hoes:** The hoe excavator has the primary cutting edge of the blade perpendicular to the axis of the handle (Fig. 7.6B). This type of instrument is used for planning tooth preparation walls and for forming line angles. It is commonly used in Class III and V preparations so as to ensure removal of any unsupported enamel prior to restoration with amalgam and composite and is especially useful in preparations to be restored with direct gold. Some sets of cutting instruments contain hoes with contra-angled shanks and long, heavy blades. These are intended for use on the prepared enamel walls of posterior teeth.

III. **Angle formers:** A special type of excavator is the angle former (Fig. 7.6C). It is used primarily for sharpening line angles and creating retentive features in dentin in preparation for gold restorations. It may also be used when placing a bevel on enamel margins. The angle former is mon-angled and has the primary cutting edge at an angle (other than 90 degrees) to the blade. It may be described as a combination of a chisel and a gingival margin trimmer and is available in pairs (right and left).

IV. **Spoons:** Spoon excavators (Fig. 7.7) are used for removing soft carious tissue and carving amalgam or direct wax patterns. The shanks may be bin-angled or triple-angled to facilitate access to various anatomical areas.

Chisels

Chisels are intended primarily for cutting enamel and may be grouped as:

I. **Straight, slightly curved or bin-angled:** The straight chisel has a straight shank and blade, with the bevel on only one side. Its primary cutting edge is perpendicular to the axis of the handle. It is similar in design to a carpenter's wood chisel (Fig. 7.9A). The shank and blade of the chisel also may be slightly curved (Wedelstaedt design) (Fig. 7.9B) or may be bin-angled (Fig. 7.9C). The force used with chisels is essentially a straight thrust or a pushing motion. A right or left type is not needed in a straight chisel because a 180-degree turn of the instrument allows for its use on either side of the preparation. The bin-angle and Wedelstaedt chisels have the primary cutting edges in a plane perpendicular to the axis of the handle and may have either a distal bevel or a mesial (reverse) bevel. The blade with a distal bevel is designed to plane a wall that faces the blade's inside surface (Fig. 7.6B and C). The blade with a mesial bevel is designed to plane a wall that faces the blade's outside surface (Fig. 7.9B and C).

II. **Enamel hatchet:** The enamel hatchet is a chisel similar in design to the ordinary hatchet except that the blade is larger, heavier and bevelled on only one side (Fig. 7.8A). It has its cutting edges in a plane that is parallel with the axis of the handle. It is used for cutting enamel and comes as right or left types for use on opposite sides of the preparation.

III. **Gingival margin trimmer:** It is similar in design to the enamel hatchet except the blade is curved and the primary cutting edge is at an angle (other than perpendicular) to the axis of the blade (Fig. 7.8B and C). It is made as right and left types. The gingival margin trimmer may be designed to afford access to mesial or distal gingival walls.

i. When the second number in the instrument formula (found on the handle) is 90–100, the pair is used on the *distal* gingival preparation wall.
ii. When the number is 75–85, the pair is used to remove unsupported enamel on the mesial gingival wall of the preparation.

The 90 and 85 pairs are for preparations of direct restorations and are used to align the gingival wall so that it is perpendicular with the external surface of the tooth structure (to ensure supported enamel). Indirect gold restorations require the placement of a distinct bevel associated with the gingival walls, and gingival margin trimmers with 100-degree and 75-degree angle pairs are designed specifically for this purpose.*Uses*:

I. The gingival margin trimmer is designed to eliminate unsupported enamel on gingival walls of proximal preparations.
II. Among other uses for these instruments is the bevelling (blunting or rounding) of the axiopulpal line angle of two-surface preparations so as to limit areas of stress concentration in subsequently placed restorations.

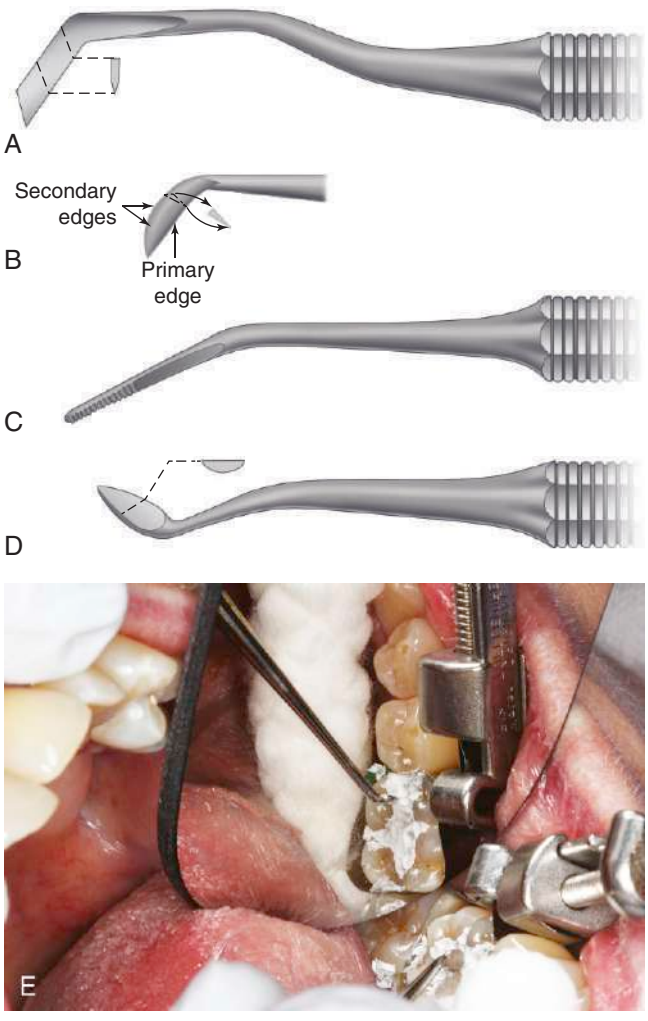


Figure 7.10 Examples of other hand instruments for cutting. (A) Finishing knife. (B) Alternative finishing knife design emphasizing secondary cutting edges. (C) Dental file. (D) Cleoid blade. (E) Discoid blade carving amalgam.

Other Cutting Instruments

Other hand-cutting instruments such as the knife, file and discoid–cleoid instrument are used for trimming restorative material rather than for cutting tooth structure.

I. **Knives** known as *finishing knives*, *amalgam knives* or *gold knives*, are designed with a thin, knife-like blade that is made in various sizes and shapes (Fig. 7.10A and B). Knives are used for trimming excess restorative material on the gingival, facial or lingual margins of a proximal restoration or trimming and contouring the surface of a Class V restoration. Sharp secondary edges on the heel aspect of the blade are useful in a scrape–pull mode.
II. **Files** (Fig. 7.10C) also may be used to trim excess restorative material and are particularly useful at gingival margins. The blades of the file are extremely thin, and the teeth of the instrument on the cutting surfaces are short and designed to make the file a push/pull instrument. Files are manufactured in various shapes and angles to enhance access to various restoration surfaces.
III. **Discoid–cleoid** (Fig. 7.10D and E) instrument is used principally for carving occlusal anatomy in unset amalgam restorations. The blades are slightly curved and the cutting edges are either circular or clawlike. The circular edge is known as a *discoid*, whereas the clawlike blade is termed *cleoid*. It also may be used to trim or burnish inlay–onlay margins. The working ends of this instrument are usually larger than the discoid end of an excavator.

Hand Instrument Techniques

Grasps

Four grasps are used with hand instruments: (i) modified pen, (ii) inverted pen, (iii) palm-and-thumb and (iv) modified palm-and-thumb. The conventional pen grasp is not an acceptable instrument grasp (Fig. 7.11A).

I. Modified Pen Grasp

The grasp that permits the greatest control of the instrument, while simultaneously maintaining ergonomic positioning of the wrist and elbow, is the modified pen grasp (Fig. 7.11B). As the name implies, it is similar but not identical to that used in holding a pen. The pads of the thumb and of the index and middle fingers contact the instrument, while the tip of the ring finger (or tips of the ring and little fingers) is placed on a nearby tooth surface of the same

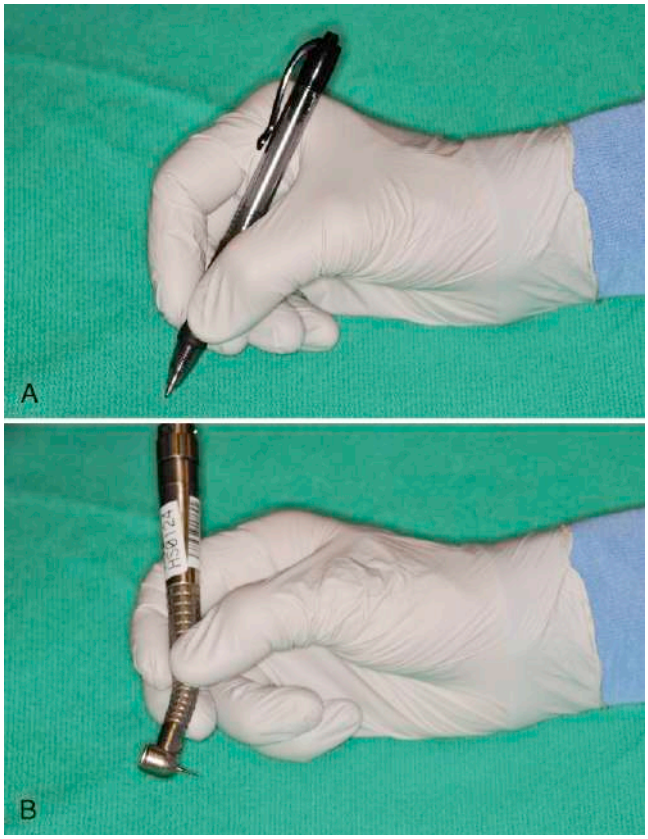


Figure 7.11 Pen grasps. (A) Conventional pen grasp. Side of middle finger is on the writing instrument. (B) Modified pen grasp. Correct position of middle finger is near the 'topside' of the instrument for good control and cutting pressure. The rest is tip (or tips) of ring finger (or ring and little fingers) on tooth (or teeth) of same arch.

arch as a rest. The palm of the hand generally is facing away from the operator. The pad of the middle finger is placed near the topside of the instrument; by this finger working with the wrist and the forearm, cutting or cleaving pressure is applied through the blade to the area requiring modification. The instrument should crossover the proximal phalange of the index finger and should not crossover at the junction area of the thumb and index finger as in the conventional pen grasp (contrast Fig. 7.11A with B). A balanced instrument design allows the application of suitable force without the instrument tending to rotate in the fingers (Fig. 7.3).

II. Inverted Pen Grasp

The finger positions of the inverted pen grasp are the same as for the modified pen grasp. The hand is rotated, however, so that the palm faces more towards the operator (Fig. 7.12). This grasp is used mostly for tooth preparations employing the lingual approach on anterior teeth.

The modified pen and inverted pen grasps are the primary grasps used in the preparation and direct restoration of teeth.



Figure 7.12 Inverted pen grasp. Palm faces more towards operator. The rest is similar to that shown for modified pen grasp (Fig. 14.11B).



Figure 7.13 Palm-and-thumb grasp. This grasp has limited use such as preparing incisal retention in a Class III preparation on a maxillary incisor. The rest is tip of thumb on tooth in same arch.

III. Palm-and-thumb Grasp

The palm-and-thumb grasp is similar to that used for holding a knife while paring an apple. The handle is placed in the palm of the hand and grasped by all the fingers, while the thumb is free of the instrument, and the rest is provided by supporting the tip of the thumb on a nearby tooth of the same arch or on a firm, stable structure. For suitable control, this grasp requires careful use during cutting. An example of an appropriate use is holding a handpiece for cutting incisal retention for a Class III preparation on a maxillary incisor (Fig. 7.13).

IV. Modified Palm-and-thumb Grasp

The modified palm-and-thumb grasp may be used when it is feasible to rest the thumb on the tooth being prepared or the adjacent tooth (Fig. 7.14). The handle of the instrument is held by all four fingers, whose pads press the handle against the distal area of the palm and the pad and first joint of the thumb. Grasping the handle under the first joints of the ring finger and little finger provides



Figure 7.14 Modified palm-and-thumb grasp. This modification allows greater ease of instrument movement and greater instrument control during pull stroke compared with palm-and-thumb grasp. The rest is tip of thumb on tooth being prepared or adjacent tooth. Note how the instrument is braced against pad and end joint of thumb.

stabilization and enables optimal control of the instrument. The modified palm-and-thumb grasp usually is employed in the area of the maxillary arch and is best adopted when the dentist is operating from a rear-chair position.

Rests

I. **Intra-arch hard tissue rest:** Proper instrument grasp must include a firm rest to steady the hand during operating procedures. When the modified pen and inverted pen grasps are used, rests are established by placing the ring finger (or both ring and little fingers) on a tooth (or teeth) of the same arch and as close to the operating site as possible (Fig. 7.12). The closer the rest areas are to the operating area, the more reliable they are. When the palm-and-thumb grasps are used, rests are created by placing the tip of the thumb on the tooth being operated on, on an adjacent tooth, or on a convenient area of the same arch (Figs. 7.13 and 7.14).

II. **Soft tissue rest:** In some instances, it is impossible to establish a rest on tooth structure, and soft tissue must be used. Neither soft tissue rests nor distant hard tissue rests afford reliable control, and they reduce the force or power that may be used safely.

III. **Indirect Rest:** Occasionally, it is impossible to establish normal finger rests with the hand holding the instrument. Under these circumstances, instrument control may be gained using the forefinger of the opposite hand on the shank of the instrument or using an indirect rest (i.e. the operating hand rests on the opposite hand, which rests on a stable oral structure).

Table 7.1 Evolution of rotary cutting equipment in dentistry

Year	Name	rpm
1728	Hand-rotated instruments	300
1871	Foot engine	700
1874	Electric engine	1,000
1914	Dental unit	5,000
1942	Diamond cutting instruments	5,000
1946	Old units converted to increased speed	10,000
1947	Tungsten carbide burs	12,000
1953	Ball-bearing handpieces	25,000
1955	Water-turbine angle handpiece	50,000
1955	Belt-driven angle handpiece (Page–Chayes)	150,000
1957	Air turbine angle handpiece	250,000
1961	Air turbine straight handpiece	25,000
1962	Experimental air-bearing handpiece	800,000
1994	Contemporary air turbine handpiece	300,000

rpm, revolutions per minute.

Contemporary Powered Cutting Equipment

Rotary-powered Cutting Equipment

Powered rotary cutting instruments, known as *dental handpieces*, are the most commonly used instruments in contemporary dentistry. Dentistry, as practiced today, would not be possible without the use of powered cutting instruments. Current dental handpieces are now highly efficient and sophisticated instruments that have evolved from their beginnings in the early 1950s. Changes in ergonomic design, weight and balance have made handpieces more comfortable to use for longer periods. Improved design helps to minimize operator arm and shoulder fatigue. Incorporation of durable fibre optics has greatly improved operator visibility. Development of light emitting diode (LED) technology has improved the quality, intensity and longevity of handpiece light sources. Modern handpieces are able to withstand the sterilization requirements of dental practice. New bearing materials and cartridges have been developed to enhance their service longevity and to contribute to noise level reduction. Handpiece chuck mechanisms have evolved such that pushbuttons, instead chuck latches or chuck tools, are used to release and change rotary instruments. Table 7.1 summarizes some significant developments of rotary dental equipment.

Two technologies are used nowadays for dental handpieces, and each has unique characteristics and benefits.



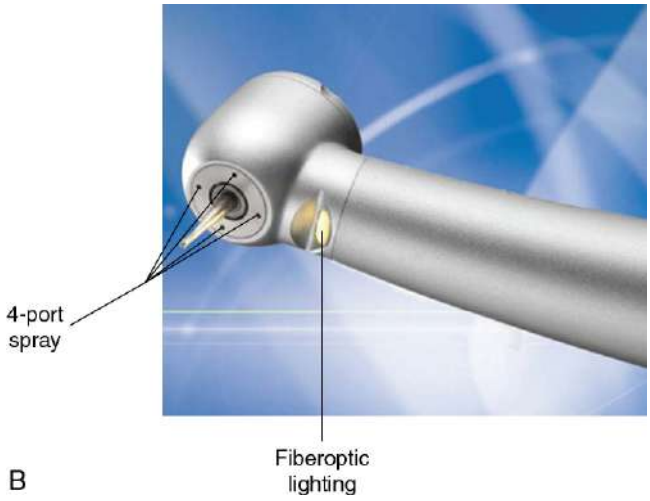


Figure 7.15 (A) Contemporary contra-angle air-turbine handpiece connected to the air–water supply line. (B) View of the handpiece showing four spray ports for cooling and fibre-optic illumination.
(Courtesy: KaVo Dental Corp., Charlotte, NC.)

I. Air-driven Handpiece

The air-driven handpiece was, for many years, the mainstay for cutting teeth in dentistry (Fig. 7.15A and B). The electric motor–driven handpiece is now becoming increasingly popular for use in all cutting applications in dentistry. The technologies for both air-driven and electric systems continue to evolve, and both systems remain very popular for everyday use in operative dentistry procedures.

Advantages:

- Air-driven systems are less costly on initial start-up and are less expensive with regard to replacing turbines compared with electric handpieces.
- Air-driven handpieces weigh less than electric handpieces, and this quality may be the most significant adjustment for clinicians who make the change from air-driven handpieces to electric handpieces.
- The size of the head of the air-driven handpiece is usually smaller.

Disadvantages:

- They create a loud, high-pitched noise that can affect the hearing of the operator and the staff over years.
- The torque and concentricity of the air-driven turbines degrade in a relatively short period, requiring replacement.
- Air-driven handpieces need turbine replacement and repairs more frequently.
- More vibration and bur chatter are associated with air-driven handpieces.



Figure 7.16 Electric handpieces and unit.
(Courtesy: DENTSPLY International, York, PA.)

II. Electric Motor-driven Handpiece

Advantages:

- They are quieter than air-driven handpieces (Fig. 7.16).
- They cut with high torque (i.e. with very little stalling).
- They offer absolute control over rotary instrument revolutions per minute (rpm).
- They offer a single motor with multiple attachments that may be used for different cutting applications. Attachments include straight type for 0.0925-in.-diameter instruments and contra-angle type for 0.0925- and 0.0626-in.-diameter instruments (see subsequent section under Shank Design).

Disadvantages:

- The initial setup expense.
- Weight and balance issues for some clinicians.

Clinical Note

The difference in the amount of cutting power is substantial in electric handpieces. Air-driven handpieces produce up to 20 W of cutting power, whereas electric handpieces are able to produce up to 60 W of cutting power. The extra cutting power in electric handpieces allows the constant torque necessary to cut various restorative materials and tooth structure regardless of the load. Unlike in the air-driven handpiece, the rotary instrument in the electric handpiece does not slow or stop (i.e. stall) as the load is increased.

Rotary Speed Ranges for Different Cutting Applications

The rotational speed of an instrument is reported in rpm. Three speed ranges are generally recognized:

- low or slow speeds (<12,000 rpm)
- medium or intermediate speeds (12,000–200,000 rpm)
- high or ultrahigh speeds (>200,000 rpm)

The terms *low speed*, *medium speed* and *high speed* are used preferentially in this textbook. Most useful instruments are rotated at either low speed or high speed. Electric handpiece motors generate up to 200,000 rpm of rotation. This speed is significantly less than the 400,000 rpm generated by air-driven handpieces. However, the electric handpiece motor has attachments with multipliers that increase rotation in ratios of 5 to 1 or 4 to 1, which enable rotational speeds effectively in the same range as air-driven handpieces.

A crucial factor is the surface speed of the instrument, that is, the velocity at which the edges of the cutting instrument pass across the surface being cut. This speed is proportional to the rotational speed and the diameter of the instrument, with large instruments having higher surface speeds at any given rate of rotation.

Low or slow speeds (<12,000 rpm): Although intact tooth structure may be removed by an instrument rotating at low speeds, it is a traumatic experience for the patient and the dentist. Low-speed cutting is ineffective, time consuming and requires a relatively heavy force application; this results in heat production at the operating site and produces vibrations of low frequency and high amplitude. Heat and vibration are the main sources of patient discomfort.[3] At low speeds, burs have a tendency to roll out of the tooth preparation and may damage the cavosurface margin or tooth surface. In addition, slow speed operation of carbide burs reduces bur life as their brittle blades are easily broken at low rpm.

Many of these disadvantages of low-speed operation do not apply when the objective is some procedure other than cutting tooth structure. The low-speed range is used for:

I. cleaning teeth,
II. caries excavation and
III. finishing and polishing procedures.

At low speeds, tactile sensation is better and generally overheating of cut surfaces is less likely. The availability of a low-speed option provides a valuable adjunct for many dental procedures.

High or ultrahigh speeds (>200,000 rpm): At high rpm, the surface speed needed for efficient cutting may be attained with smaller and more versatile cutting instruments. This speed is used for tooth preparation and removing old restorations. Other high-rpm advantages include

I. Diamond and carbide cutting instruments remove tooth structure faster and with less pressure, vibration and heat generation.
II. The types of rotary cutting instruments needed is reduced because smaller sizes are more universal in application.
III. The operator has better control and greater ease of operation.
IV. Instruments last longer.
V. Patients are generally less apprehensive because annoying vibrations and operating time are decreased.
VI. Increased operation efficiency results in the opportunity to treat several teeth in the same arch at the same appointment.

Variable control of rpm makes the handpiece more versatile. This feature allows the operator to easily obtain the optimal speed for the size and type of rotating instrument at any stage of a specific operation. Air-driven handpiece rpm may be controlled, but precise control is more difficult since the operator's pressure on the foot-operated rheostat controls the speed of the handpiece. All electric handpieces have an electronic control module that, in concert with an adjustable rheostat, allows precise control of maximum rpm for specific procedural situations.

For infection control, all dental handpieces are now sterilized. Continual sterilization will produce degradation in clinical performance (longevity, power, turbine speed, fibre-optic transmission, eccentricity, noise, chuck performance, water spray pattern).[4] Most handpieces require lubrication after sterilization, and excess oil may be present during the start-up operation and increase risk of contamination of the preparation. Several companies offer automated equipment to precisely clean and lubricate the handpiece after each use. It is recommended to run the handpiece for a few seconds before initiating dental procedures in which the deposition of oil spray onto tooth structure might interfere with processes such as dental adhesion.

Air Abrasion Equipment

Concept: Air abrasion techniques rely on the transfer of kinetic energy from a stream of powder particles on the surface of tooth structure or a restoration to produce a fractured surface layer, resulting in roughness for bonding or disruption for cutting.

Clinical applications:

Alternative methods of cutting enamel and dentin have been assessed periodically. In the mid-1950s, air-abrasive cutting was tested, but several clinical problems precluded general acceptance. Most importantly, no tactile sense was associated with air-abrasive cutting of tooth structure. This made it difficult for the operator to determine the cutting progress within the tooth preparation. Additionally the abrasive dust interfered with visibility of the cutting site and tended to mechanically etch the surface of the dental mirror. Preventing the patient or office personnel from inhaling abrasive dust posed an additional difficulty.

Contemporary air abrasion equipment is helpful for stain removal, debriding pits and fissures before sealing and micromechanical roughening of surfaces to be bonded (enamel, cast metal alloys or porcelain) (Fig. 7.17A).[7] This approach works well when organic material is being removed and when only a limited amount of enamel or dentin is involved. Although promoted for caries excavation, air abrasion is not able to produce well-defined preparation wall and margin details that are possible with conventional rotary cutting techniques. Generally, the finest stream of abrading particles still generates an effective cutting width that is far greater than the width of luted cement margins or the errors tolerable in most caries excavations. Air abrasion does allow the roughening of surfaces to be bonded, luted or repaired and this may be accomplished intra-orally or extra-orally. Roughening by air abrasion by itself is not a substitute for acid-etching techniques. Roughening improves bonding. Acid etching alone or after roughening, however, always produces a better bond than air abrasion alone.[8]

The energy transfer event is affected by many things, including powder flow rate, particle size, pressure, angulation, surface composition and clearance angle variables (Fig. 7.17B). The most common error made by operators of air abrasion units is holding the tip at the wrong distance from the surface for the desired action. Greater distances significantly reduce the energy of the stream.[9] Short distances may produce unwanted cutting actions, such as when only surface stain removal is being attempted. The potential for unwanted cutting is a significant problem when employing an air-polishing device (e.g. Prophy Jet) to clean the surfaces of dentin and enamel.[10–13] When used properly, however, units designed for air polishing tooth surfaces may be quite effective and efficient (Fig. 7.17C).

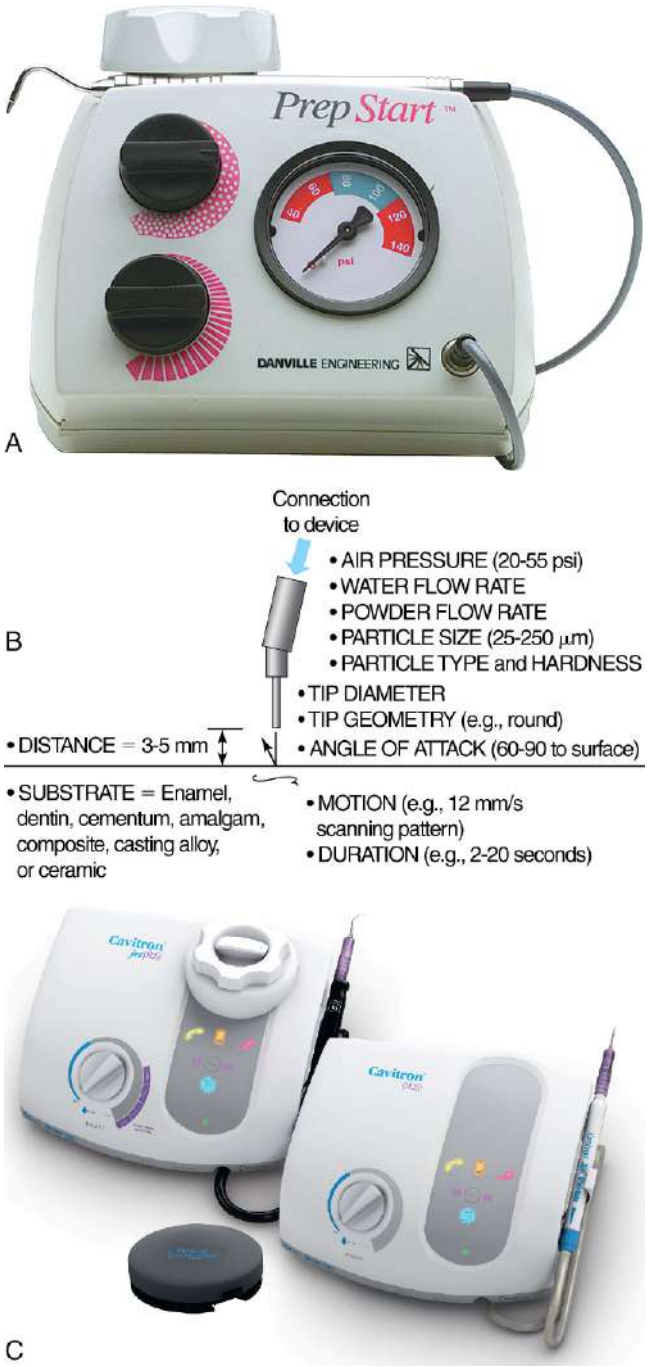


Figure 7.17 (A) Example of contemporary air abrasion unit for removal of superficial enamel defects or stains, debriding pits and fissures for sealant application, or roughening surfaces to be bonded or luted. (B) Schematic representation of range of variables associated with any type of air abrasion equipment. The cleaning or cutting action is a function of kinetic energy imparted to the actual surface, which is affected by variables concerning the particle size, air pressure, angulation with surface, type of substrate and method of clearance. (C) Example of air abrasion equipment used for tooth cleaning showing the Prophy tip and handle attached by a flexible cord to the control unit with the reservoir of powder and source of water (*left*).

Part A, Courtesy: Danville Materials, Inc., San Ramon, CA; part B, Courtesy: Kunselman B.: *Effect of air-polishing shield on the abrasion of PMMA and dentine* (Master's thesis), Chapel Hill, NC, 1999, School of Dentistry, University of North Carolina; part C, Courtesy: Dentsply International, York, PA.)

Laser Equipment

Lasers are devices that produce beams of coherent and very-high-intensity light. Numerous current and potential uses of lasers in dentistry have been identified that involve the treatment of soft tissues and the modification of hard tooth structures.[5,6] The word LASER is an acronym for *'light amplification by stimulated emission of radiation'*. A crystal or gas is excited to emit photons of a characteristic wavelength that are amplified and filtered to make a coherent light beam. The effects of the laser depend on the power of the beam and the extent to which the beam is absorbed.

Box 7.2 Common laser types used in dentistry[30]

Laser type construction	Wavelength(s)
Argon gas laser	488 and 515 nm
KTP (Potassium-titanyl-phosphate) solid state	532 nm
Helium–neon gas laser	633 nm
Diode semiconductor	635, 670, 810, 830 and 980 nm
Nd:YAG (neodymium-doped yttrium aluminum garnet) solid state	1064 nm
Er,Cr:YSGG (Erbium, chromium-doped yttrium, scandium, gallium and garnet) solid state	2780 nm
Er:YAG (Erbium yttrium aluminum garnet crystal) solid state	2940 nm
Er:YSGG solid state	2790 nm
CO_2 gas laser	9600 and 10600 nm

Lasers are unique as they produce a *collimated* and *coherent* beam of light (coherent: waves of one wavelength all in phase). This is quite different from the incoherent light of the sun. The almost parallel, single-wavelength light rays that make up the collimated laser beam has a considerably higher power density, and can be focused to a much smaller spot size than the randomly radiated rays of the sun. Consequently, a much more efficient power density is achieved with a laser. The various kinds of lasers that are potentially employed in dentistry is given in Box 7.2 [30]

Hard Tissue Lasers

The lasers in the middle-infrared region have been employed for cavity preparation and caries removal. The lasers commonly recommended for these applications are:

I. Er:YAG of 2940 nm
II. Er:YSGG of 2790 nm
III. Er,Cr:YSGG of 2780 nm

These infrared wavelengths correspond to the peak absorption range of water in the infrared spectrum. As the hard tissue cutting efficiency of these lasers relies on water-based absorption, the efficiency of *ablation* is greatest for Er:YAG laser. Apart from caries removal and tooth preparation the other hard tissue procedures where lasers have been employed include:

- Desensitization of cervical dentin
- Laser analgesia
- Laser-enhanced fluoride uptake

Clinical Limitations

- Current laser units are relatively expensive than air-driven and electric motor cutting instruments.
- These units are used primarily for either soft tissue applications and have few applications for hard tissue surface modification.
- They may be used for tooth preparations; however, generation of a defined margin or tooth preparation surface is not possible. Lasers are an inefficient means by which to remove large amounts of enamel or dentin and have the potential to generate unwanted amounts of heat.
- They are not able to remove existing amalgam or ceramic dental restorations.
- No single laser type is suitable for all potential laser applications. Lasers may never replace the high-speed dental handpiece.
- Currently, available laser instruments have proven to be relatively inefficient and impractical for tooth preparation and have not achieved widespread popularity.

Rotary Cutting Instruments

The individual instruments intended for use with dental handpieces are manufactured in hundreds of sizes, shapes and types. This variation is in part a result of the need for specialized designs for particular clinical applications or to fit particular handpieces, but much of the variation also results from individual preferences on the part of dentists. Since the introduction of high-speed techniques in clinical practice, a rapid evolution of technique and an accompanying proliferation of new instrument designs have occurred. Nevertheless, the number of instruments essential for use with any one type of handpiece is comparatively small, especially in the case of high-speed handpieces.

Common Design Characteristics

Despite the great variation among rotary cutting instruments, they share certain design features. Each instrument consists of three parts (Fig. 7.18):

I. shank,
II. neck and
III. head.

Each part has its own function, influencing its design and the materials used for its construction. The term *shank* has different meanings as applied to rotary instruments and to hand instruments.

Shank Design

The *shank* is the part that fits into the handpiece, accepts the rotary motion from the handpiece and provides a bearing surface to control the alignment and concentricity of the instrument. The shank design and dimensions vary with the handpiece for which it is intended. The American Dental Association (ADA) Specification No. 23 for dental excavating burs includes five classes of instrument shanks.[14] Three of these (Fig. 7.19) are commonly encountered.



Figure 7.18 Normal designation of three parts of rotary cutting instruments.

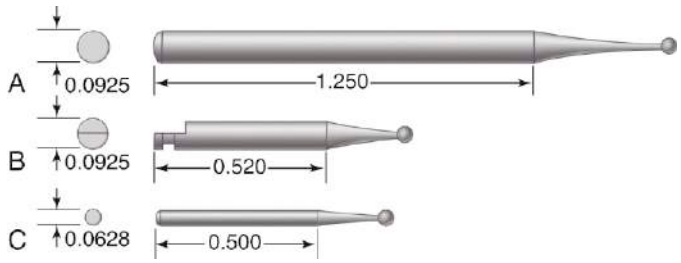


Figure 7.19 Characteristics and typical dimensions (in inches) of three common instrument shank designs for straight handpiece (A), right angle latch-type handpiece (B) and friction-grip contra-angle type handpiece (C).

I. ***Straight handpiece shank:*** The shank portion of the straight handpiece instrument is a simple cylinder that is held in the handpiece by a metal chuck. Straight handpiece instruments are now rarely used for preparing teeth but are used for oral surgery extraction procedures. They are commonly used, however, for external finishing and polishing indirect restorations.

II. ***Right angle latch handpiece shank:*** The more complicated shape of the right angle latch-type shank reflects the different mechanisms by which these instruments are held in the handpiece. Their shorter overall length substantially improves access to posterior regions of the mouth compared with straight handpiece instruments. Handpieces that use right angle latch-type burs normally have a metal bur tube within which the instruments fit as closely as possible, while still permitting easy interchange. The posterior portion of the shank is flattened on one side so that the end of the instrument fits into a D-shaped socket at the bottom of the bur tube. Engagement in this area is required for bur rotation to occur. Latch-type instruments are not retained in the handpiece by a chuck, but by a retaining latch that slides into the groove found at the shank end of the instrument. This type of instrument is used predominantly at low-speed and medium-speed ranges for caries excavation and finishing procedures. At these speeds, the small amount of potential wobble inherent in the clearance between the rotary instrument and the handpiece bur tube is controlled by the lateral pressure exerted during cutting procedures. At higher speeds, the latch-type shank/bur tube design is inadequate to provide required alignment and concentricity for the rotary instrument head; as a result, an improved shank/clutch design is required for these speeds.

III. ***Friction-grip handpiece shank:*** The friction grip shank design was developed for use with high-speed handpieces. This design is smaller in overall length than the latch-type instruments, providing a further improvement in access to the posterior regions of the mouth. The shank is a simple cylinder manufactured to close dimensional tolerances. As the name implies, friction-grip instruments originally were designed to be held in the handpiece by friction between the shank and a plastic or metal chuck. Newer handpiece designs have metal chucks that close to make a positive contact with the bur shank. Careful dimensional control of the shanks of high-speed instruments is important because even minor variations in shank diameter can cause substantial variation in instrument performance and problems with insertion, retention and removal.

Neck Design

As shown in Fig. 7.18, the *neck* is the intermediate portion of an instrument that connects the head to the shank. Except in the case of the larger, more massive instruments, the neck normally tapers from the shank diameter to a smaller size immediately adjacent to the head. The main function of the neck is to transmit rotational and translational forces to the head. At the same time, it is desirable for the operator to have the greatest possible visibility of the cutting blades (or diamond particles) on the head and the greatest manipulative freedom. For this reason, the neck dimensions represent a compromise between the need for a large cross section to provide strength and a small cross section to improve access and visibility.

Head Design

The *head* is the working part of the instrument, the cutting edges or points that perform the desired shaping of tooth structure. The shape of the head and the material used to construct it are closely related to its intended application and technique of use. The heads of instruments show greater variation in design and construction than either of the other main components of the rotary cutting instrument. For this reason, the characteristics of the head form the basis on which rotary instruments are usually classified.

Many characteristics of the heads of rotary instruments could be used for classification. Most important among these is the division into:

I. Bladed instruments (dental burs) and
II. Abrasive instruments (diamond abrasive instruments).

Material of construction, head size and head shape are additional characteristics that are useful for further subdivision. Bladed and abrasive instruments exhibit substantially different clinical performance, even when operated under nearly identical conditions. This appears to result from differences in the mechanism of cutting that are inherent to their design. The two main divisions that will be utilized for this discussion are *dental burs* and *diamond abrasive instruments*.

Dental Burs

The term *bur* is applied to all rotary cutting instruments that have bladed cutting heads. This includes instruments intended for finishing metal restorations, surgical removal of bone and tooth preparation.

Historical Development of Dental Burs

Steel burs: The earliest burs were handmade. They were not only expensive but also variable in dimension and performance. The shapes, dimensions and nomenclature of modern burs are directly related to those of the first machine-made burs introduced in 1891.[15] Early burs were made of steel. Steel burs perform well, cutting human dentin at low speeds, but dull rapidly at higher speeds or when cutting enamel. When burs are dulled, the reduced effectiveness in cutting creates increased heat and vibration.

Carbide burs: These were introduced in 1947, and have largely replaced steel burs for tooth preparation. Steel burs now are used mainly for finishing procedures. Carbide burs perform better than steel burs at all speeds, and their superiority is greatest at high speeds. All carbide burs have heads of cemented carbide in which microscopic carbide particles, usually tungsten carbide, are held together in a matrix of cobalt or nickel. Carbide is much harder than steel and less prone to dull during cutting.

In most burs, the carbide head is attached to a steel neck (and shank) by welding or brazing. The substitution of steel for carbide in the portions of the bur where greater wear resistance is not required has certain advantages: it permits the manufacturer more freedom of design in attaining the characteristics desired in the instrument and allows economy in the cost of materials of construction.

Although most carbide burs have the joint located in the posterior part of the head, others are sold that have the joint located within the shank and have carbide necks and heads. Carbide is stiffer and stronger than steel, but it is also more brittle. A carbide neck subjected to a sudden blow or shock will fracture, whereas a steel neck will bend. A bur that is even slightly bent produces increased vibration and overcutting as a result of increased runout (see upcoming description of runout). Although steel necks reduce the risk of fracture during use, they may cause severe problems if bent. Either type may be satisfactory, and other design factors are varied to take maximal advantage of the properties of the material used.

Bur Classification Systems

To facilitate the description, selection and manufacture of burs, it is highly desirable to have some agreed-on shorthand designation, which represents all variables of a particular head design by some simple code. In the United States, dental burs traditionally have been described in terms of an arbitrary numerical code for head size and shape (e.g. 2 = 1-mm-diameter round bur; 57 = 1-mm-diameter straight fissure bur; 34 = 0.8-mm-diameter inverted cone bur).[16] Despite the complexity of the system, it is still in common use. Other countries similarly developed and used arbitrary systems. Newer classification systems such as those developed by the International Dental Federation (*Federation Dentaire Internationale*) and International Standards Organization (ISO) tend to use separate designations for shape (usually a shape name) and size (usually a number giving the head diameter in tenths of a millimetre) (e.g. round 010; straight fissure plain 010 and inverted cone 008).[17,18]

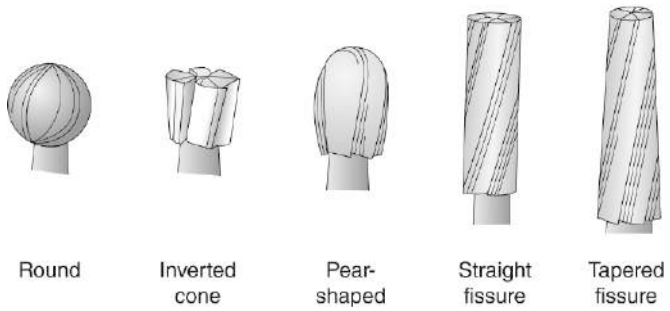


Figure 7.20 Basic bur head shapes.
(From Finkbeiner BL, Johnson CS: *Mosby's comprehensive dental assisting*, St. Louis, MO, 1995, Mosby.)

Shapes

The term *bur shape* refers to the contour or silhouette of the head. The basic head shapes are round, inverted cone, pear, straight fissure and tapered fissure (Fig. 7.20).

Round bur is spherical. This shape customarily has been used for initial entry into the tooth, extension of the preparation, preparation of retention features and caries removal.

Inverted cone bur is a portion of a rapidly tapered cone with the apex of the cone directed towards the bur shank. Head length is approximately the same as the head diameter. This shape is particularly suitable for providing undercuts in tooth preparations.

Pear-shaped bur is a portion of a slightly tapered cone with the small end of the cone directed towards the bur shank. The end of the head either is continuously curved (Fig. 7.20) or is flat with rounded corners where the sides and flat end intersect (Fig. 7.21A; 245). The No. 245 is an elongated pear bur (length three times the width) and is advocated for tooth preparations for amalgam.

Straight fissure bur is an elongated cylinder. Some dentists advocate this shape for amalgam tooth preparation. Modified burs of this design with slightly curved tip angles are available.

Tapered fissure bur is a portion of a slightly tapered cone with the small end of the cone directed away from the bur shank. This shape is used for tooth preparations for indirect restorations, for which freedom from undercuts is essential for successful withdrawal of patterns and final seating of the restorations. Tapered fissure burs may also be designed with a flat end with the tip corners slightly rounded.

Among these basic shapes, variations are possible. Fissure and inverted-cone burs may have half-round or domed ends. Taper and cone angles may vary. The ratio of head length to diameter may be varied. In addition to



Figure 7.21 Burs used in recommended procedures. Bur sizes ¼, ½, 2, 4, 33½ and 169L are standard carbide burs available from various sources. The 245 and 271 burs are nonstandard carbide burs that do not conform to the current American Dental Association (ADA) standard numbering system. They are designed to combine rounded corners with flat ends and are available from several manufacturers. The diamond instruments shown are wheel (*x*), flame (*y*) and tapered cylinder (*z*). Various twist drills are illustrated. Particular drills often are provided as specified by manufacturers of pin-retention systems.

shape, other features may be varied, such as the number of blades, spiral versus axial patterns for blades and continuous versus cross-cut blade edges (Figs. 7.21 and 7.22).

Sizes

In the United States, the number designating bur size also traditionally has served as a code for head design. This numbering system for burs was originated by the S.S. White Dental Manufacturing Company in 1891 for their first machine-made burs. It was extensive and logical, so other domestic manufacturers found it convenient to adopt it for their burs as well. As a result, for more than 60 years, a general uniformity existed for bur numbers in the United States.

The original numbering system grouped burs by 9 shapes and 11 sizes. The ½ and ¼ designations (both very small round burs) were added later when smaller instruments were included in the system. All original bur designs had continuous blade edges. Later, when cross-cut burs were found to be more effective for cutting dentin at low speeds, cross-cut versions of many bur sizes were introduced. This modification was indicated by adding 500 to the number of the equivalent non-cross-cut size (e.g. a No. 57 with cross-cut was designated No. 557). Similarly, a 900 prefix was used to indicate a head design intended for end cutting only (e.g. except for differences in blade design, a No. 957, No. 557 and No. 57 bur all had the same head dimensions). These changes occurred gradually over time without disrupting the system. The sizes in common use in 1955 are shown in Table 7.2. The system changed rapidly thereafter, but where the numbers are still used, the designs and dimensions remain the same.

Modifications in Bur Design

As available handpiece speeds increased after 1950, particularly after the high-speed turbine handpieces were

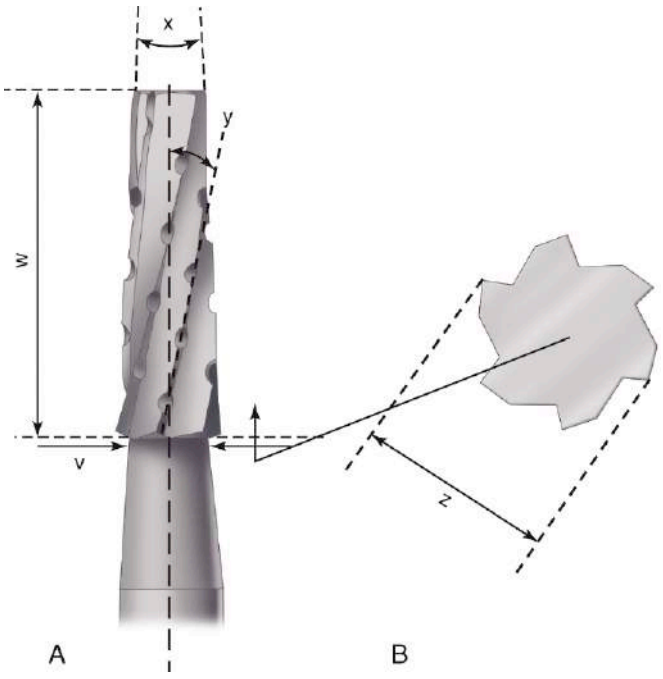


Figure 7.22 Design features of bur heads (illustrated using No. 701 bur cross-cut). (A) Lateral view—neck diameter (*v*), head length (*w*), taper angle (*x*) and spiral angle (*y*). (B) End view—head diameter (*z*).

introduced, a new wave of modification of bur sizes and shapes occurred. Numerous other categories arose as new variations in blade number or design were created. Some of the numbers assigned to the burs were selected arbitrarily. With the introduction of new bur sizes and elimination of older sizes, much of the logic in the system has no longer been maintained, and many dentists and manufacturers no longer recognize the original significance of the numbers used for burs. The number of standard sizes that have continued in use has been reduced. This has been most obvious in the decreased popularity of large-diameter burs. The cutting effectiveness of carbide burs is greatly increased at high speeds.[19] This is particularly true of the small-diameter sizes, which did not have sufficient peripheral speed for efficient cutting when used at lower rates of rotation. As the effectiveness of small burs has increased, they have replaced larger burs in many procedures. Three other major trends in bur design are discernible:

I. *Reduced use of cross-cuts*: Cross-cuts are needed on fissure burs to obtain adequate cutting effectiveness at low speeds, but they are not needed at high speeds. Because cross-cut burs used at high speeds tend to produce unduly rough surfaces, many of the cross-cut sizes originally developed for low-speed use have been replaced by non-cross-cut instruments of the same dimension for high-speed use.[20] In many instances, the non-cross-cut equivalents were available; a No. 57 bur might be used at high speed, whereas a No. 557 bur was preferred for low-speed use. Non-cross-cut versions of the 700 series burs have become popular, but their introduction precipitated a crisis in the bur numbering system because no number traditionally had been assigned to burs of this type.

II. *Extended heads on fissure burs*: Carbide fissure burs have been introduced with head lengths extended two to three times those of the normal tapered fissure burs of similar diameter. Such a design would never have been practical using a brittle material such as carbide if the bur were to be used at low speed. The applied force required to make a bur cut at speeds of 5000–6000 rpm would normally be sufficient to fracture such an

Table 7.2 Standard bur head sizes—carbide and steel (1955–Present)

	Head diameters in inches (mm*)													
	0.020	0.025	0.032	0.040	0.048	0.056	0.064	0.073	0.082	0.091	0.100	0.110	0.120	0.130
Head shapes	(0.5)	(0.6)	(0.8)	(1.0)	(1.2)	(1.4)	(1.6)	(1.9)	(2.1)	(2.3)	(2.5)	(2.8)	(3.0)	(3.3)
Round	¼	½	1	2	3	4	5	6	7	8	9	10	11	–
Wheel	–	11½	12	–	14	–	16	–	–	–	–	–	–	–
Inverted cone	–	33½	34	35	36	37	38	39	40	–	–	–	–	–
Plain fissure	–	55½	56	57	58	59	60	61	62	–	–	–	–	–
Round cross-cut	–	–	–	502	503	504	505	506	–	–	–	–	–	–
Straight fissure cross-cut	–	–	556	557	558	559	560	561	562	563	–	–	–	–
Tapered fissure cross-cut	–	–	–	700	701	–	702	–	703	–	–	–	–	–
End cutting fissure	–	–	–	957	958	959	–	–	–	–	–	–	–	–

Note: Nonstandard burs are not shown in this table.
*Millimetre values rounded to the nearest 0.1 mm.

attenuated head. The extremely light applied pressures needed for cutting at high speed permit many modifications of burs, however, that would have been impractical at low speed.

III. *Rounding of sharp tip angles*: The third major trend in bur design has been towards rounding of the sharp tip corners. Early contributions to this trend were made by Markley and Sockwell.[8] Sharp angles produced in tooth preparations by conventional burs result in areas of high-stress concentration with increased potential for tooth fracture. Bur heads with rounded corners allow better distribution of occlusal stress in restored teeth, help retain the strength of the tooth by preserving vital dentin and facilitate the adaptation of restorative materials. Carbide burs and diamond instruments with this design last longer because no sharp corners prone to chipping and wear are present. Burs with rounded transitions facilitate tooth preparation with desired features of flat preparation floors and rounded internal line angles.

Many of these new and modified bur designs simplify the techniques and reduce the effort needed for optimal results. Although the development of new bur sizes and shapes has greatly increased the number of different types in current use, the number actually required for clinical effectiveness has been reduced. Most instruments recommended in this text for the preparation of teeth are illustrated in Fig. 7.21. The selection includes standard head designs and modified designs of the types just discussed. Included in Table 7.3 are the significant head dimensions of these standard and modified burs.

A problem related to the dimensions and designations of rotary dental instruments worldwide arose because each country developed its own system of classification. The rapid rate at which new bur designs were introduced during the transition to high-speed techniques threatened to cause a complete breakdown in the numbering system. As various manufacturers simultaneously developed and marketed new burs of similar design, the risk of similar burs being given different numbers or different burs being given the same number increased.

Progress towards the development of an international numbering system for basic bur shapes and sizes under the auspices of the ISO has been slow. For other design features, the trend instead seems to be towards the use of individual manufacturer's code numbers. Throughout the remaining text, the traditional US numbers are used, where possible. The few exceptions are shown in Fig. 7.21 and Table 7.3.

Additional Features in Head Design

Numerous factors other than head size and shape are involved in determining the clinical effectiveness of a bur design.[21,22] Fig. 7.22 shows a lateral view and a cross-sectional view of a No. 701 cross-cut tapered fissure bur in which several of these factors are illustrated. The lateral view (Fig. 7.22A) shows *neck diameter*, *head diameter*, *head*

Table 7.3 Names and key dimensions of recommended burs

Manufacturer's size number	ADA size number	ISO size number	Head diameter (mm)	Head length (mm)	Taper angle (degrees)	Shape
¼	¼	005	0.50	0.40	–	Round
½	½	006	0.60	0.48	–	Round
2	2	010	1.00	0.80	–	Round
4	4	014	1.40	1.10	–	Round
33½	33½	006	0.60	0.45	12	Inverted cone
169	169	009	0.90	4.30	6	Tapered fissure
169L*	169L	009	0.90	5.60	4	Elongated tapered fissure
329	329	007	0.70	0.85	8	Pear, normal length
330	330	008	0.80	1.00	8	Pear, normal length
245**,***	330L	008	0.80	3.00	4	Pear, elongated
271**	171	012	1.20	4.00	6	Tapered fissure
272**	172	016	1.60	5.00	6	Tapered fissure

ADA, American Dental Association; *ISO*, International Standard Organization.
*Similar to the No. 169 bur except for greater head length.
**These burs differ from the equivalent ADA size by being flat ended with rounded corners. The manufacturer's number has been changed to indicate this difference.
***Similar to the No. 330 bur except for greater head length.

length, *taper angle*, *blade spiral angle* and *cross-cut size* and *spacing* as they apply to this bur size.

Of these features, *head length* and *taper angle* are primarily descriptive and may be varied within limits consistent with the intended use of the bur. This bur originally was designed for use at low speeds in preparing teeth for cast restorations. The taper angle is intended to approximate the desired occlusal divergence of the lateral walls of the preparations, and the head length must be long enough to reach the full depth of the normal preparation. These factors do not otherwise affect the performance of the bur.

Neck diameter is important functionally because a neck that is too small results in a weak instrument unable to resist lateral forces. Too large a neck diameter may interfere with visibility and use of the part of the bur head next to the neck and may restrict access for coolants. As the head of a bur increases in length or diameter, the moment arm exerted by lateral forces increases, and the neck needs to be larger.

Compared with these factors, two other design variables, the *spiral angle* and *cross-cutting*, have considerably greater influence on bur performance. There is a tendency towards reduced spiral angles on burs intended exclusively for high-speed operation in which a large spiral angle (which would produce a smoother preparation) is not needed. The smaller spiral angle produces more efficient cutting.

As noted previously, cross-cut bur designs have notches in the blade edges to increase cutting effectiveness at low and medium speeds. A minimum amount of perpendicular force is required to make a blade grasp the surface and initiate cutting as it passes across the surface. The harder the surface, the duller the blade; and the greater its length, the more is the force required to initiate cutting. By reducing the total length of bur blade that is actively cutting at any one time, the cross-cuts effectively increase the cutting pressure resulting from rotation of the bur and the perpendicular pressure holding the blade edge against the tooth.

As each cross-cut blade cuts, it leaves small ridges of tooth structure standing behind the notches. Because the notches in two succeeding blades do not line up with each other, the ridges left by one blade are removed by the following one at low or medium speed. At the high speed attained with air-driven handpieces, however, the contact of the bur with the tooth is not continuous, and usually only one blade cuts effectively.[23] Under these circumstances, although the high cutting rate of cross-cut burs is maintained, the ridges are not removed and a much rougher cut surface results.[20]

A cross-sectional view of the No. 701 bur is shown in Fig. 7.22B. This cross section is made at the point of largest head diameter and is drawn as seen from the shank end. The bur has six blades uniformly spaced with depressed areas between them. These depressed areas are properly known as *flutes*. The number of blades on a bur is always even because even numbers are easier to produce in the manufacturing process, and instruments with odd numbers of blades cut no better than those with even numbers.

The number of blades on an excavating (cutting) bur may vary from 6 to 8 to 10. Burs intended mainly for finishing procedures usually have 12–40 blades. The greater the number of blades, the smoother is the cutting action at low speeds. Most burs are made with at least six blades because they may need to be used in this speed range. In the high-speed range, no more than one blade seems to cut effectively at any one time, and the remaining blades are, in effect, spares. The tendency for the bur to cut on a single blade is often a result of factors other than the bur itself. Nevertheless, it is important that the bur head be as symmetrical as possible. Two terms are in common use to measure this characteristic of bur heads: *concentricity* and *runout*.

I. *Concentricity* is a direct measurement of the symmetry of the bur head itself. It measures how closely a single circle can be passed through the edges of all of the blades. Concentricity is an indication of whether one blade is longer or shorter than the others. It is a static measurement not directly related to function.
II. *Runout* is a dynamic test measuring the accuracy with which all blade edges pass through a single point when the instrument is rotated. It measures not only the concentricity of the head but also the accuracy with which the centre of rotation passes through the centre of the head. Even a perfectly concentric head exhibits substantial runout if the head is not centred precisely on the long axis of the bur, the bur neck is bent, the bur is not held straight in the handpiece chuck, or the chuck is eccentric relative to the handpiece bearings. The runout can never be less than the concentricity, and it is usually substantially greater.

Clinical Note

Runout is the more significant term clinically because it is the primary cause of vibration during cutting and is the factor that determines the minimum diameter of the hole that may be prepared by a given bur. Burs normally cut holes measurably larger than the head diameter because of runout defects.

Bur Blade Design

The actual cutting action of a bur (or a diamond abrasive) occurs in a very small region at the edge of the blade (or at the point of a diamond chip). In the high-speed range, this effective portion of the individual blade is limited to no more than a few thousandths of a centimetre adjacent to the blade edge. Fig. 7.23 is an enlarged schematic view of this portion of a bur blade. Several terms used in the discussion of blade design are illustrated.

Each blade has two sides—the *rake face* (towards the direction of cutting) and the *clearance face*—and three

important angles—the *rake angle*, the *edge angle* and the *clearance angle*. The optimal angles depend on factors such as the mechanical properties of the blade material, the mechanical properties of the material being cut, the rotational speed and diameter of the bur and the lateral force applied by the operator through the handpiece to the bur.

Rake angle: The rake angle is the most important design characteristic of a bur blade. For cutting hard, brittle materials, a negative rake angle minimizes fractures of the cutting edge, increasing the bur life. A rake angle is said to be negative when the rake face is ahead of the radius (from cutting edge to axis of bur), as illustrated in Fig. 7.23.

Edge angle: Increasing the edge angle reinforces the cutting edge and reduces the likelihood for the edge of the blade to fracture. Carbide bur blades have higher hardness and are more wear resistant, but they are more brittle than steel blades and require greater edge angles to minimize fractures. The three angles cannot be varied independently of each other. An increase in the clearance angle causes a decrease in the edge angle.

Clearance angle: The clearance angle eliminates rubbing friction of the clearance face, provides a stop to prevent the bur edge from digging into the tooth structure excessively and reduces the radius of the blade back of the cutting edge to provide adequate flute space or clearance space for the cutting debris that accumulates ahead of the following blade.

Carbide burs normally have blades with slight negative rake angles and edge angles of approximately 90 degrees. Their clearance faces either are curved or have two surfaces to provide a low clearance angle near the edge and a greater clearance space ahead of the following blade.

Diamond Abrasive Instruments

The second major category of rotary dental cutting instruments involves abrasive cutting rather than blade cutting. Abrasive instruments are based on small, angular particles of a hard substance held in a matrix of softer material.

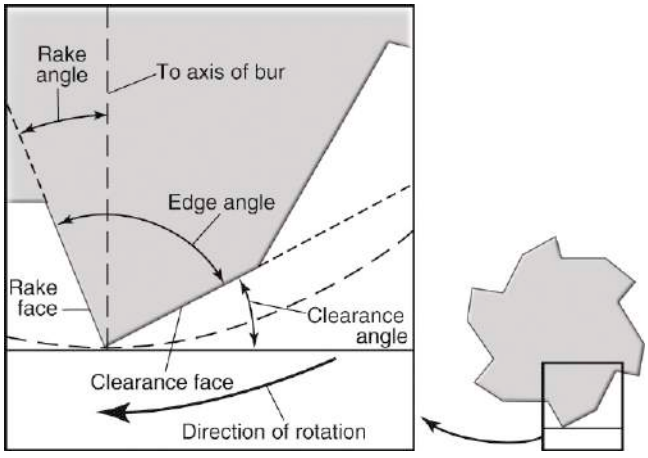


Figure 7.23 Bur blade design. Schematic cross section viewed from shank end of head to show rake angle, edge angle and clearance angle.

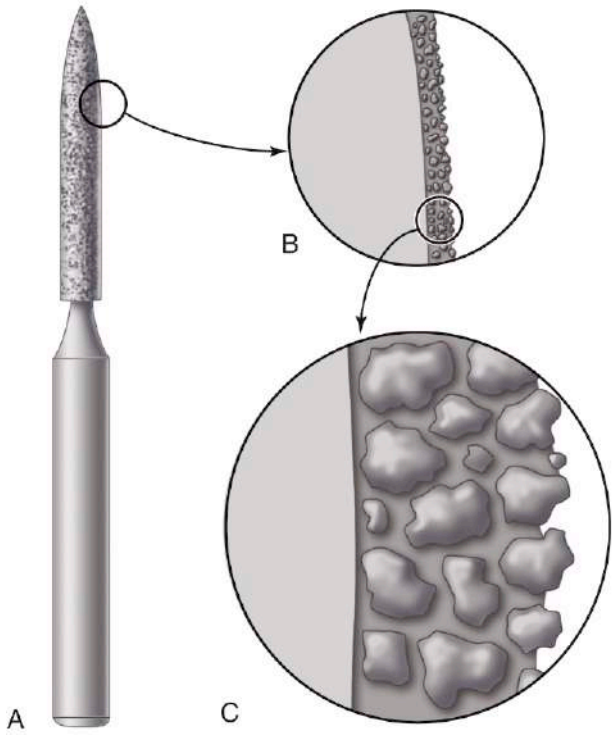


Figure 7.24 Diamond instrument construction. (A) Overall view. (B) Detail of abrasive layer. (C) Detail of particle bonding.

Cutting occurs at numerous points where individual hard particles protrude from the matrix rather than along a continuous blade edge. This design causes definite differences in the mechanisms by which the two types of instruments cut and in the applications for which they are best suited.

Abrasive instruments are generally grouped as diamond or other instruments. Diamond instruments have had great clinical impact because of their long life and great effectiveness in cutting enamel and dentin. Diamond instruments for dental use were introduced in the United States in 1942 at a time before carbide burs were available and when interest in increased rotational speeds was beginning to expose the limitations of steel burs. The earliest diamond instruments were substitutes for previously used abrasive points of other types used for grinding and finishing.[24] The vastly superior performance in these applications led to their immediate acceptance. The shortage of burs, as a result of wartime demands of the 1940s, forced the use of diamond instruments. Increased usage revealed their durability in cutting enamel and promoted the development of operative techniques employing them.

Terminology

Diamond instruments consist of three parts:

I. metal blank,
II. powdered diamond abrasive and
III. metallic bonding material that holds the diamond powder onto the blank (Fig. 7.24).

The blank in many ways resembles a bur without blades. It has the same essential parts: head, neck and shank. The shank dimensions, similar to those for bur shanks, depend on the intended handpiece. The neck is normally a tapered section of reduced diameter that connects the shank to the head, but for large disk-shaped or wheel-shaped instruments, it may be the same diameter as the shank. The head of the blank is undersized compared with the desired final dimensions of the instrument, but its size and shape determine the size and shape of the finished instrument. Dimensions of the head make allowance for a fairly uniform thickness of diamonds and bonding material on all sides. Some abrasive instruments are designed as a mandrel with a detachable head, which is much more practical for abrasive disks that have very short lifetimes.

The diamonds employed are industrial diamonds, either natural or synthetic, that have been crushed to powder then carefully graded for size and quality. The shape of the individual particle is important because of its effect on the cutting efficiency and durability of the instrument, but the careful control of particle size is probably of greater importance. The diamonds generally are attached to the blank by electroplating a layer of metal on the blank while holding the diamonds in place against it. Although the electroplating holds the diamonds in place, it also tends to cover much of the diamond surfaces. Various proprietary techniques allow greater diamond exposure and resultant increased cutting effectiveness.

Table 7.4 Standard categories of shapes and sizes for diamond cutting instruments

Head shapes	Profile variations
Round	–
Football	Pointed
Barrel	–
Cylinder	Flat-, bevel-, round- or safe-end
Inverted cone	–
Taper	Flat-, round- or safe-end
Flame	–
Curettage	–
Pear	–
Needle	'Christmas tree'
Interproximal	Occlusal anatomy
Donut	–
Wheel	–

Classification

Diamond instruments currently are marketed in a myriad of head shapes and sizes (Table 7.4) and in all of the standard shank designs. Most of the diamond shapes parallel those for burs (Fig. 7.25). This great diversity arose in part as a result of the relative simplicity of the manufacturing process. Because it is possible to make diamond instruments in almost any shape for which a blank can be manufactured, they are produced in many highly specialized shapes on which it would be impractical to place cutting blades. This has been a major factor in establishing clinical uses for these unique shapes, which are not in direct competition with burs.

Head Shapes and Sizes

Diamond instruments are available in a wide variety of shapes and in sizes that correspond to all except the smallest diameter burs. The greatest difference lies in the

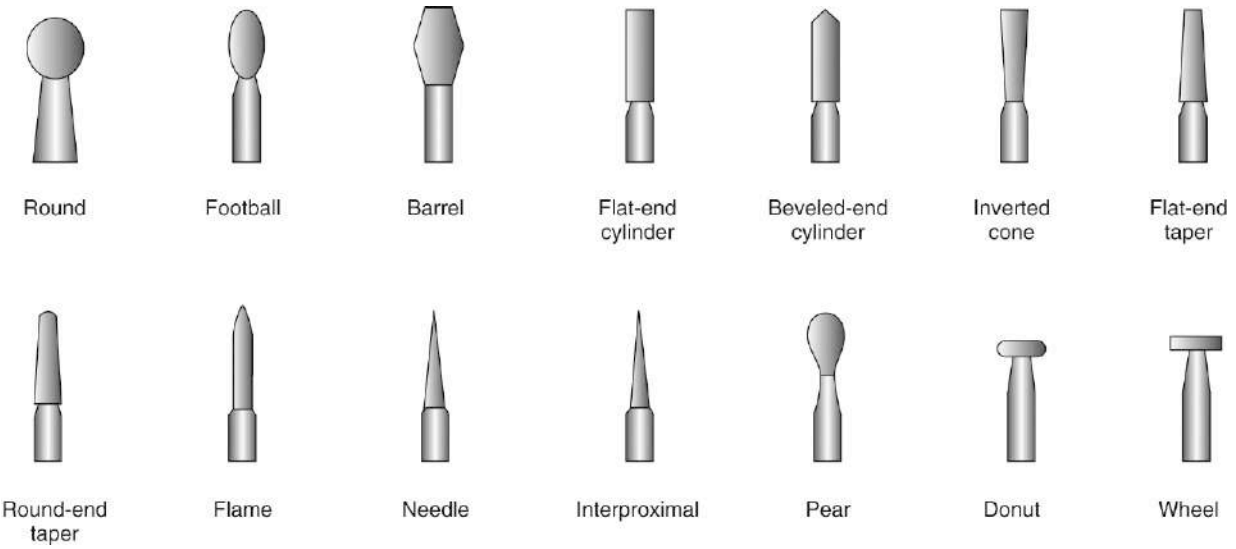


Figure 7.25 Characteristic shapes and designs for a range of diamond cutting instruments.

diversity of other shapes and sizes in which diamond instruments are produced. Even with many subdivisions, the size range within each group is large compared with that found among the burs. More than 200 shapes and sizes of diamonds are currently marketed.

Because of their design with an abrasive layer over an underlying blank, the smallest diamond instruments cannot be as small in diameter as the smallest burs, but a wide range of sizes is available for each shape. No one manufacturer produces all sizes, but each usually offers an assortment of instruments that includes popular shapes and sizes. Because of the lack of uniform nomenclature for diamond instruments, it is often necessary to select them by inspection to obtain the desired shape and size. It is essential to indicate the manufacturer when attempting to describe diamond instruments by catalogue number.

Diamond Particle Factors

The clinical performance of diamond abrasive instruments depends on the size, spacing, uniformity, exposure and bonding of the diamond particles. Increased pressure causes the particles to dig into the surface more deeply, leaving deeper scratches and removing more tooth structure.

Diamond particle size is commonly categorized for diamond preparation instruments as one of the following[24]:

I. coarse (125–150 μm),
II. medium (88–125 μm),
III. fine (60–74 μm) and
IV. very fine (38–44 μm).

These ranges correspond to standard sieve sizes for separating particles. When using large particle sizes, the number of abrasive particles that can be placed on a given area of the head is decreased. For any given force that the operator applies, the pressure on each particle tip is greater. The resulting pressure is also increased if diamond particles are more widely spaced so that fewer are in contact with the surface at any one time. The final clinical performance of diamond instruments is strongly affected by the technique used to take advantage of the design factors for each instrument.

Diamond finishing instruments use even finer diamonds (10–38 μm) to produce relatively smooth surfaces for final finishing with diamond polishing pastes. Surface finishes of less than 1 μm are considered clinically and may be routinely attained by using a series of progressively finer polishing steps.

Proper diamond instrument speed and pressure are the major factors in determining service life.[25] Properly used diamond instruments last almost indefinitely. A primary cause of failure of diamond instruments is loss of the diamonds from critical areas. This loss results from the use of excess pressure in an attempt to increase the cutting rate at inadequate speeds.[26]

Other Abrasives: Finishing and Polishing Instruments

In addition to diamond instruments, many other types of abrasive instruments are used in dentistry. At one time they were extensively used for tooth preparation, but their use is now primarily restricted to shaping, finishing and polishing restorations in the clinic and in the laboratory.

Classification

In these instruments, as in the diamond instruments, the cutting surfaces of the head are composed of abrasive particles held in a continuous matrix of softer material. They may be divided into two distinct groups—*moulded instruments* and *coated instruments*. Each uses various abrasives and matrix materials.

I. *Moulded abrasive instruments*: They have heads that are manufactured by moulding or pressing a uniform mixture of abrasive and matrix around the roughened end of the shank or cementing a pre-moulded head to the shank. In contrast to diamond instruments, moulded instruments have a much softer matrix and therefore wear during use. The abrasive is distributed throughout the matrix so that new particles are exposed by the wear. These instruments are made in a full range of shapes and sizes. The mounted heads are often termed *points* and *stones*. Hard and rigid moulded instrument heads use rigid polymer or ceramic materials for their matrix and are commonly used for grinding and shaping procedures. Other moulded instrument heads use flexible matrix materials, such as rubber, to hold the abrasive particles. These are used predominantly for finishing and polishing procedures. Moulded unmounted disks or wheelstones are attached by a screw to a mandrel of suitable size for a given handpiece that has a threaded hole in the end. This design permits the instruments to be changed easily and enable extended use of the mandrel.
II. *Coated abrasive instruments*: They are mostly disks that have a thin layer of abrasive cemented to a flexible backing. This construction allows the instrument to conform to the surface contour of a tooth or restoration. Most flexible disks are designed for reversible attachment to a mandrel. Coated abrasive instruments may be used in the finishing and smoothing procedures of certain enamel walls (and margins) of tooth preparations for indirect restorations but are most often used in finishing procedures for restorations.

The abrasives are softer and are less wear resistant than diamond powder; as a result, they tend to lose their sharp edges and cutting efficiency with use and are then discarded. In contrast, moulded instruments are intended to partially regenerate through the gradual loss of their worn

outer layers but may require that the operator reshape them to improve their concentricity or usefulness in finishing various anatomic areas. Reshaping is accomplished by applying a truing or shaping stone against the rotating instrument.

Materials

The matrix materials usually are phenolic resins or rubber. Some moulded points may be sintered, but most are resin bonded. A rubber matrix is used primarily to obtain a flexible head on instruments to be used for polishing. A harder, non-flexible rubber matrix is often used for moulded silicon carbide (SiC) disks. The matrix of coated instruments is usually one of the phenolic resins.

Synthetic or natural abrasives may be used, including silicon carbide, aluminum oxide, garnet, quartz, pumice and cuttlebone. The hardness of the abrasive has a major effect on the cutting efficiency. The Moh's hardness values for important dental abrasives are shown in Table 7.5.

I. *Silicon carbide* (*SiC*): SiC usually is used in moulded rounds, tree or bud shapes, wheels and cylinders of various sizes. These points are normally gray-green, available in various textures, usually fast cutting (except on enamel) and produce a moderately smooth surface. Moulded unmounted disks are black or a dark colour, have a soft matrix, wear more rapidly than stones and produce a moderately rough surface texture. These disks are termed *carborundum disks* or *separating disks*.
II. *Aluminium oxide* is used for the same instrument designs as those for silicon carbide disks. Points are usually white, rigid, fine textured and less porous and produce a smoother surface than SiC.
III. *Garnet* (*reddish*) and *quartz* (*white*) are used for coated disks that are available in a series of particle sizes and range from coarse to medium-fine for use in initial finishing. These abrasives are hard enough to cut tooth structure and all restorative materials, with the exception of some ceramics.
IV. *Pumice* is a powdered abrasive produced by crushing foamed volcanic glass into thin glass flakes. The flakes cut effectively, but they break down rapidly. Pumice is used with rubber disks and wheels, usually for initial polishing procedures.
V. *Cuttlebone* is derived from the cuttlefish, a relative of squid and octopus. It is becoming scarce and gradually is being replaced by synthetic substitutes. It is a soft white abrasive, used only in coated disks for final finishing and polishing. It is soft enough that it reduces the risk of unintentional damage to tooth structure during the final stages of finishing.

Cutting Mechanisms

For cutting, it is necessary to apply sufficient pressure to make the cutting edge of a blade or abrasive particle dig into the surface. Local fracture occurs more easily if the strain rate is high (high rotary instrument surface speed) because the surface that is being cut responds in a brittle fashion. The process by which rotary instruments cut tooth structure is complex and not fully understood. The following discussion addresses cutting evaluations,

Table 7.5 Hardness values of restorative materials, tooth structure and abrasives

	Knoop hardness	Brinell hardness	Moh's hardness
Dentin	68	48	3–4
Enamel	343	300	5
Dental composite	41–80	60–80	5–7
Dental amalgam	110	–	4–5
Gold alloy (type III)	–	110	–
Feldspathic porcelain	460	–	6–7
Pumice	–	–	6
Cuttlebone	–	–	7
Garnet	–	–	6.5–7
Quartz	800	600	7
Aluminum oxide	1500	1200	9
Silicon carbide	2500	–	9.5
Diamond	>7000	>5000	10

cutting instrument design, proposed cutting mechanisms and clinical recommendations for cutting.

Evaluation of Cutting

Cutting may be measured in terms of effectiveness and efficiency. Certain factors influence one but not the other.[27]

- *Cutting effectiveness* is the rate of tooth structure removal (mm/min or mg/s).
- *Cutting efficiency* is the percentage of energy actually producing the cutting. Cutting efficiency is reduced when energy is wasted as heat or noise.

Effectiveness does not consider potential side effects such as heat or noise. It is possible to increase effectiveness while decreasing efficiency. A dull bur may be made to cut faster than a sharp bur by applying a greater pressure, but experience indicates that this results in a great increase in heat production and reduced efficiency.[28]

It is generally agreed that increased rotational speed results in increased effectiveness and efficiency. Adverse effects associated with increased speeds are heat, vibration and noise. Heat has been identified as a primary cause of pulpal injury. Air–water sprays do not prevent the production of heat, but do serve to remove it before it causes a damaging increase in temperature within the tooth.

Bladed Cutting

The following discussion focuses on rotary bladed instruments but also is applicable to bladed hand instruments. Tooth structure, similar to other materials, undergoes brittle and ductile fracture.

- *Brittle fracture* is associated with crack production usually by tensile forces.
- *Ductile fracture* involves plastic deformation of material usually proceeding by shear forces.

Extensive plastic deformation may also produce local work hardening and encourage brittle fracture. Low-speed cutting tends to proceed by plastic deformation before tooth structure fracture. High-speed cutting, especially of enamel, proceeds by brittle fracture.

The rate of stress application (or strain rate) affects the resultant properties of materials. In general, the faster the rate of loading, the greater are the strength, hardness, modulus of elasticity and brittleness of a material. A cutting instrument with a large diameter and high rotational speed produces a high surface speed and a high stress (or strain) rate.

Many factors interact to determine which cutting mechanism is active in a particular situation[19,29]:

- Mechanical properties of tooth structure
- Design of the cutting edge or point
- Linear speed of the instrument's surface
- Contact force applied
- Power output characteristics of the handpiece

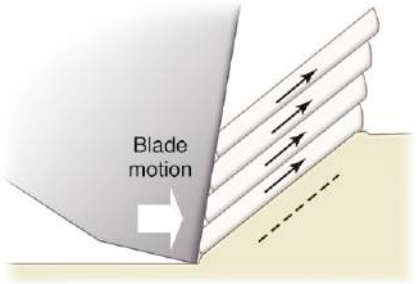


Figure 7.26 Schematic representation of bur blade (*end view*) cutting a ductile material by shearing mechanism. Energy is required to deform the material removed and produce new surface.

For the blade to initiate the cutting action, it must be sharp, must have a higher hardness and modulus of elasticity than the material being cut and must be pressed against the surface with sufficient force. The high hardness and modulus of elasticity are essential to concentrate the applied force on a small enough area to exceed the shear strength of the material being cut. As shown in Fig. 7.26, sheared segments accumulate in a distorted layer that slides up along the rake face of the blade until it breaks or until the blade disengages from the surface as it rotates. These chips accumulate in the clearance space between blades until washed out or thrown out by centrifugal force.

Mechanical distortion of tooth structure ahead of the blade produces heat. Frictional heat is produced by the rubbing action of the cut debris against the rake face of the blade and the blade tip against the cut surface of the tooth immediately behind the edge. This can produce extreme temperature increases in the tooth and the bur in the absence of adequate cooling. The transfer of heat is not instantaneous, and the reduced temperature increase observed in teeth cut at very high speeds may be caused, in part, by the removal of the heated surface layer of the tooth structure by a following blade before the heat can be conducted into the tooth.

Abrasive Cutting

The following discussion is pertinent to all abrasive cutting situations, but diamond instruments are used as the primary example.[12] The cutting action of diamond abrasive instruments is similar in many ways to that of bladed instruments, but key differences result from the properties, size and distribution of the abrasive. The very high hardness of diamonds provides superior resistance to wear. A diamond instrument that is not abused has little or no tendency to dull with use. Individual diamond particles have very sharp edges, are randomly oriented on the surface and tend to have large negative rake angles.

Diamond instruments cutting ductile materials (dentin): When diamond instruments are used to cut ductile materials,

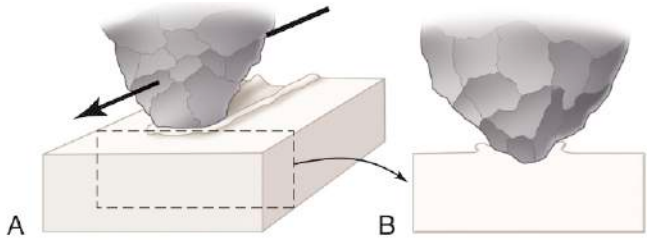


Figure 7.27 Schematic representation of an abrasive particle cutting ductile material. (A) Lateral view. (B) Cross-sectional view. Material is displaced laterally by passage of an abrasive particle, work hardened and subsequently removed by other particles.

some material is removed as debris, but much material flows laterally around the cutting point and is left as a ridge of deformed material on the surface (Fig. 7.27). Repeated deformation work hardens the distorted material until irregular portions become brittle, break off and are removed. *This type of cutting is less efficient than that by a blade; burs are generally preferred for cutting ductile materials such as dentin*.

Diamond instruments cutting brittle materials (enamel): Diamonds cut brittle materials by a different mechanism. Most cutting results from tensile fractures that produce a series of subsurface cracks (Fig. 7.28). *Diamonds are most efficient when used to cut brittle materials and are superior to burs for the removal of dental enamel*. Diamond abrasives are commonly used for milling in computer-aided design/computer-assisted manufacturing (CAD/CAM) or copy-milling applications.

Cutting Recommendations

Overall, the requirements for effective and efficient cutting include:

I. using a contra-angle handpiece, air–water spray for cooling,
II. high operating speed (>200,000 rpm),
III. light pressure and
IV. a carbide bur or diamond instrument.

Carbide burs are better for end cutting, produce lower heat and have more blade edges per diameter for cutting. They are used effectively for punch cuts to enter tooth structure,

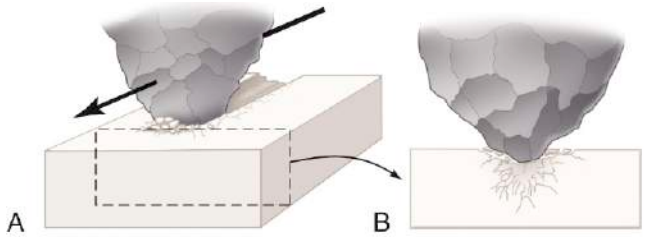


Figure 7.28 Schematic representation of abrasive particle cutting brittle material. (A) Lateral view. (B) Cross-sectional view. Subsurface cracks caused by the passage of abrasive particles intersect, undermining small pieces of material, which are removed easily by following abrasive particles.

intra-coronal tooth preparation, amalgam removal, small preparations and secondary retention features.

Diamond instruments have higher hardness, and coarse diamonds have high cutting effectiveness. Diamonds are more effective than burs for intra-coronal and extra-coronal tooth preparations, bevelling enamel margins on tooth preparations and enameloplasty.

Hazards with Cutting Instruments

Almost everything done in a dental office involves some risk to the patient, the dentist or the dental assistant. For the patient, pulpal dangers arise from tooth preparation and restoration procedures. Soft tissue dangers are also present. Everyone is potentially susceptible to eye, ear and inhalation dangers. Careful adherence to normal precautions can, however, eliminate or minimize most risks associated with the use of cutting instruments.

Pulpal Precautions

The use of cutting instruments may harm the pulp by exposure to mechanical vibration, heat generation, desiccation and loss of dentinal tubule fluid or transection of odontoblastic processes. As the thickness of remaining dentin decreases, the pulpal insult (and response) from heat or desiccation increases. Slight to moderate injury produces a localized, protective pulpal response in the region of the cut tubules. In severe injury, destruction extends beyond the cut tubules, often resulting in pulpal abscess and death of the pulp. These pulpal sequelae (recovery or necrosis) take from 2 weeks to 6 months or longer to become apparent, depending on the extent and degree of the trauma. Although a young pulp is more prone to injury, it also recovers more effectively compared with an older pulp, in which the recuperative powers are slower and less effective.

Enamel and dentin are good thermal insulators and protect the pulp if the quantity of heat is not too great and the remaining thickness of tissue is adequate. The longer the time of cutting and the higher the local temperature produced, the greater is the threat of thermal trauma. *The remaining tissue is effective in protecting the pulp in proportion to the square of its thickness*. Steel burs produce more heat than carbide burs because of inefficient cutting. Burs and diamond instruments that are dull or plugged with debris do not cut efficiently, resulting in heat production. When used without coolants, diamond instruments generate more damaging heat compared with carbide burs.

The most common instrument coolants are *air* and *air–water sprays*. Air alone as a coolant is not effective in preventing pulpal damage because it needlessly desiccates dentin and damages odontoblasts. Air has a much lower heat capacity than water and is much less efficient in absorbing unwanted heat. An air coolant alone should be used only when visibility is a problem, such as during the finishing procedures of tooth preparations. At such times, air coolant combined with lower rotational speed, light

and intermittent instrument application should be used to enhance vision and minimize trauma.

Air–water spray is universally used to cool, moisten and clear the operating site during normal cutting procedures. In addition, the spray lubricates, cleans and cools the cutting instrument, increasing its efficiency and service life. A well-designed and properly directed air–water spray also helps keep the gingival crevice open for better visualization when gingival extension is necessary. The use of a water spray and its removal by effective high-volume evacuation (HVE) are especially important when old amalgam restorations are removed. HVE helps in rapid, efficient removal of released mercury vapour and aids in increased operator visibility.

During normal cutting procedures, a layer of debris, described as a *smear layer*, is created that covers the cut surfaces of the enamel and dentin. The smear layer on dentin is moderately protective because it occludes the dentinal tubules and inhibits the outward flow of tubular fluid and the inward penetration of microleakage contaminants. The smear layer is still porous, however. When air alone is applied to dentin, local desiccation may produce fluid flow and affect the physiologic status of the odontoblastic processes in the underlying dentin. Air is applied only to the extent of removing excess moisture, leaving a glistening surface.

Soft Tissue Precautions

The lips, tongue and cheeks of the patient are the most frequent areas of soft tissue injury. The handpiece should never be operated unless good access to and visualization of the cutting site are available. A rubber dam is helpful in isolating the operating site. When the dam is not used, the dental assistant may retract the soft tissue on one side with a mouth mirror, cotton roll or evacuator tip. The dentist usually is able to manage the other side with a mirror or cotton roll, or both. If the dentist must work alone, the patient may help by holding a retraction-type saliva ejector evacuator tip after it is positioned in the mouth.

The rotating instrument in a handpiece does not stop immediately when the foot control is released. The operator must wait for the instrument to stop or be extremely careful when removing the handpiece from the mouth so as not to contact and lacerate soft tissues. The large disk is one of the most dangerous instruments used in the mouth. Such disks are seldom indicated intra-orally. They should be used with light, intermittent application and with extreme caution.

With high-speed and low-speed air-driven handpieces, sluggish handpiece performance will alert the dental practitioner to maintenance issues such as a dull bur or worn or clogged gears or bearings. A poorly maintained electric handpiece does not provide a similar warning that maintenance is needed. Instead, if an electric handpiece is worn out, damaged or clogged, the electric motor sends increased power to the handpiece head or attachment in order to maintain handpiece performance. This increased power can rapidly generate heat at the head of the handpiece attachment. Because the heat buildup is so rapid and is efficiently conducted through the metal handpiece, a burn occurring in the patient may be the first indication of handpiece problems. Patients have been severely burned when electric handpieces have overheated during dental procedures (Fig. 7.29). Some patients have suffered third-degree burns that required reconstructive surgery. Burns may not be apparent to the operator or the patient until after the tissue damage has occurred because the anesthetized patient is not able to feel the tissue burning and the handpiece housing insulates the operator from the heated attachment. Adhering to strict maintenance guidelines recommended by the manufacturers is critical to prevent overheating in electric handpieces. The clinician must be aware that improperly maintained, damaged or worn-out devices have the potential to overheat without warning.



Figure 7.29 This patient suffered burn from the overheated bearing of an electric handpiece. Because the patient was anesthetized, he was unaware of the burn as it occurred from the overheated handpiece.

The dentist and assistant must always be aware of the patient's response during the cutting procedures. A sudden reflex movement caused by patient gagging, swallowing or coughing has the potential to result in serious injury from a rotating cutting instrument. If an accident does occur and soft tissue is damaged, the operator should remain calm and control any haemorrhage with a pressure pack. The patient should be told what has happened, and medical assistance should be obtained, if needed.

Eye Precautions

The operator, the assistant and the patient should wear glasses with side shields to prevent eye damage from airborne particles during operative procedures using rotary instrumentation. When high speeds are used, particles of old restorations, tooth structure, bacteria and other debris are discharged at high speeds from the patient's mouth. Sufficiently strong HVE applied by the dental

assistant near the operating site helps limit this problem. Protective glasses are always indicated when rotary instrumentation is being used. The dentist, being in the direct path of such particles, is more likely to receive injury than the assistant or the patient. If an eye is injured, it should be covered by a clean gauze pad until medical attention is obtained.

In addition to routine airborne debris, airborne particles may be produced occasionally by matrix failure of moulded abrasive cutting instruments. Hard matrix wheels may crack or shatter into relatively large pieces. Soft abrasive wheels or points may increase in temperature during use, causing the rubber matrix to debond explosively from the abrasive into fine particles.

Precautions must be taken to prevent eye injury from unusual light sources such as visible light–curing units and laser equipment. Dental personnel and patients should be protected from high-intensity visible light with the use of coloured plastic shields (attached to the fibre-optic tip). Laser light can be inadvertently reflected from many surfaces in the dental operatory; the operatory should be closed, and everyone should wear protective goggles (see earlier Section Laser Equipment).

Ear Precautions

Various sounds are known to affect people in different ways. Soft music or random sounds such as rainfall usually have a relaxing or sedative effect. Loud noises are generally annoying and may contribute to mental and physical distress. A noisy environment decreases the ability to concentrate, increases risk of accidents and reduces overall efficiency. Extremely loud noises such as explosions or continuous exposure to high noise levels can cause permanent damage to the hearing mechanism.

An objectionable high-pitched whine is produced by some air-driven handpieces at high speeds. Aside from the annoying aspect of this noise, hearing loss could result from continued exposure. Potential damage to hearing from noise depends on:

I. the intensity or loudness (db),
II. frequency (cps) of the noise,
III. duration (time) of the noise and
IV. susceptibility of the individual.

Older age, existing ear damage, disease and medications are other factors that can accelerate hearing loss. Normal ears require that the intensity of sound reach a certain minimal level before the ear can detect it. This is known as *auditory threshold* and may vary with the frequency and exposure to other sounds. When subjected to a loud noise of short duration, a protective mechanism of the ear causes it to lose some sensitivity temporarily. This is described as *temporary threshold shift*. If sufficient time is allowed between exposures, complete recovery occurs. Extended or continuous exposure is much more likely to result in a permanent threshold shift with persistent hearing loss. The loss may be caused by all frequencies, but often high-frequency sounds affect hearing more severely. A certain amount of unnoticed noise (ambient noise level) is present even in a quiet room (20–40 db). An ordinary conversation averages 50–70 db in a frequency range of 500–2500 cps.

Air-driven handpieces with ball bearings, free running at 30-lb air pressure, may have noise levels of 70–94 db at high frequencies. Noise levels greater than 75 db in frequency ranges of 1000–8000 cps may cause hearing damage. Noise levels vary among handpieces produced by the same manufacturer. Handpiece wear and eccentric rotating instruments may cause increased noise. *Protective measures are recommended when the noise level reaches 85 db with frequency ranges of 300–4800 cps*.

Protection is mandatory in areas where the level transiently reaches 95 db. The effect of excessive noise levels depends on exposure times. Normal use of a dental handpiece is one of intermittent application that generally is less than 30 min per day and represents little risk of developing hearing loss over time. Earplugs may be used to reduce the level of exposure, but these have several obvious drawbacks. Room soundproofing helps limit sound reflection and may be accomplished with absorbing materials used on ceilings, walls and floors. Antinoise devices may also be used to cancel unwanted sounds.

Inhalation Precautions

Aerosols and vapours are created by cutting tooth structure and restorative materials and are a health hazard to all present. Aerosols are fine dispersions in air of water, tooth debris, microorganisms and/or restorative materials. Cutting amalgam or composite resin produces submicron particles and vapour. The particles that may be inadvertently inhaled have the potential to produce alveolar irritation and tissue reactions. Vapour from cutting amalgam is predominantly mercury and should be captured and removed, as much as possible, by HVE near the tooth being operated on. The vapours generated during cutting or polishing by thermal decomposition of polymeric restorative materials (sealants, acrylic resin, composites) are predominantly resin monomers. The resin monomers may be captured and efficiently removed by careful HVE during the cutting or polishing procedures.

A rubber dam protects the patient against oral inhalation of aerosols or vapours, but nasal inhalation of vapour and finer aerosol may still occur. Disposable masks worn by dental office personnel filter out bacteria and all but the finest particulate matter. Masks do not, however, filter out mercury or resin monomer vapours. The biologic effects of mercury hazards and appropriate office hygiene measures are discussed in Chapter 12.

Summary

Modern dental equipment allows efficient removal and shaping of tooth structures and restorative materials. Hand instruments and rotary-powered cutting burs and abrasive instruments remain essential components of the dental armamentarium. A wide variety of strategic shapes and sizes of instruments are available for use in patient care. Proper understanding and use of hand instruments, hand pieces, burs and abrasives enable optimal accomplishment of dental procedures with minimal risk to the patient and dental team.

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